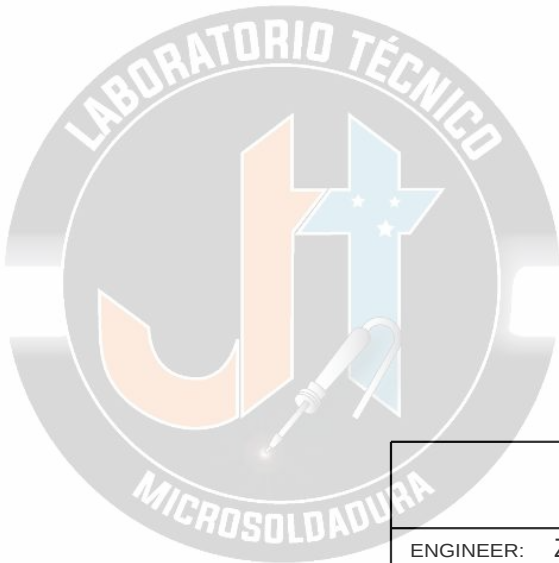


RESERVE 1400-1449

- 1. ADD THIRD MIC ON MAIN CONNECTOR, BUT NOT COMPATIBLE WITH RIGA M
- 2. ADDED 2ST CONNECTOR FOR FRONT CAMMERA,NEED CHANG NET
- 3. DELETE PD
- 4. ADD FAULT_N ON PMI632 PIN 44
- 5. OPTION RESISTER SHOULD BE 240K



APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: ZHANG BEI		BLOCK TITLE:	
LAST MODIFIED BY: ZHANG BEI Wed Mar 06 16:56:43 2019		defender_sm6150	
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		CAD FILE NAME:	
		PAGE 1 OF 69	

H

G

F

E

D

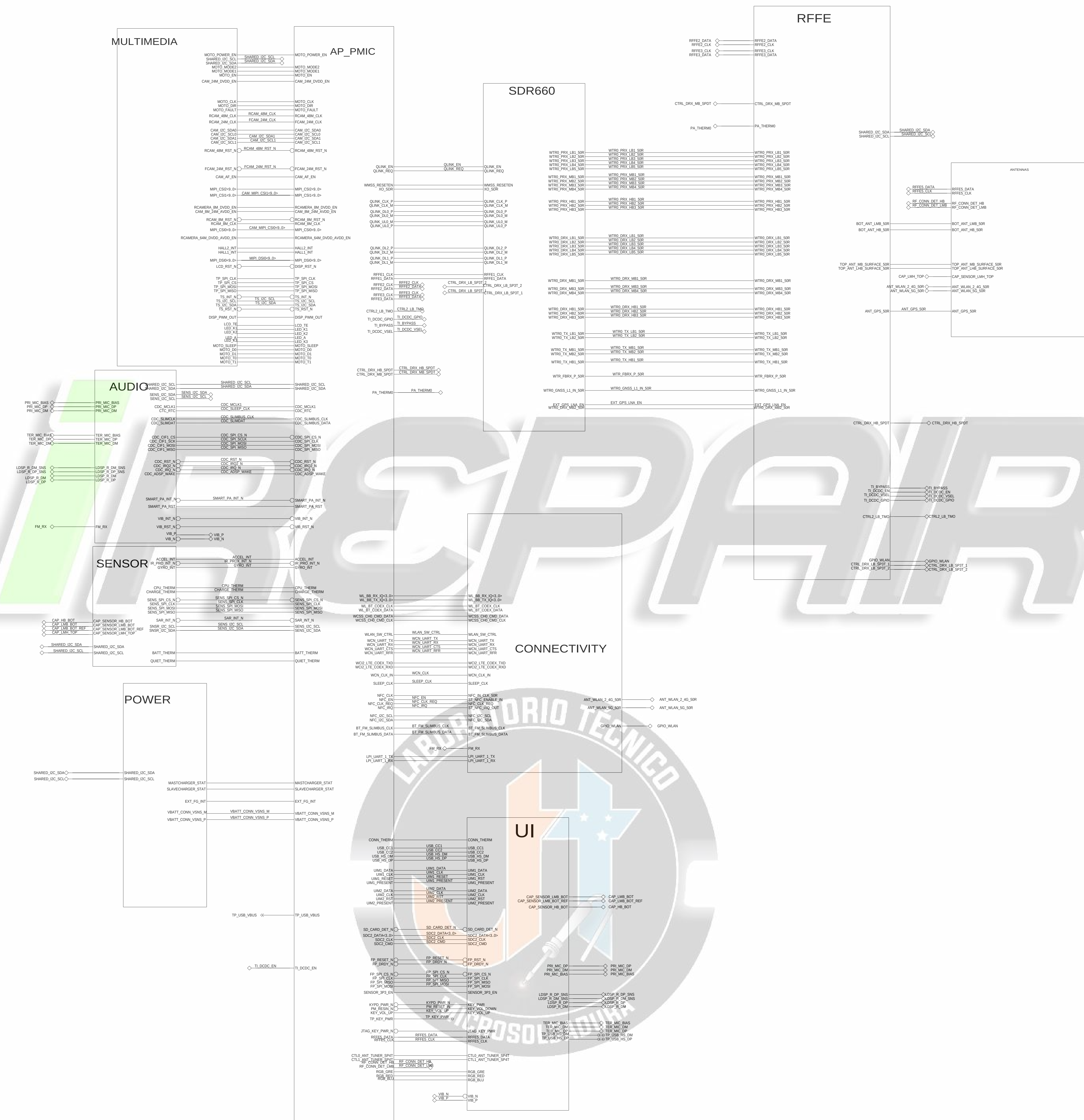
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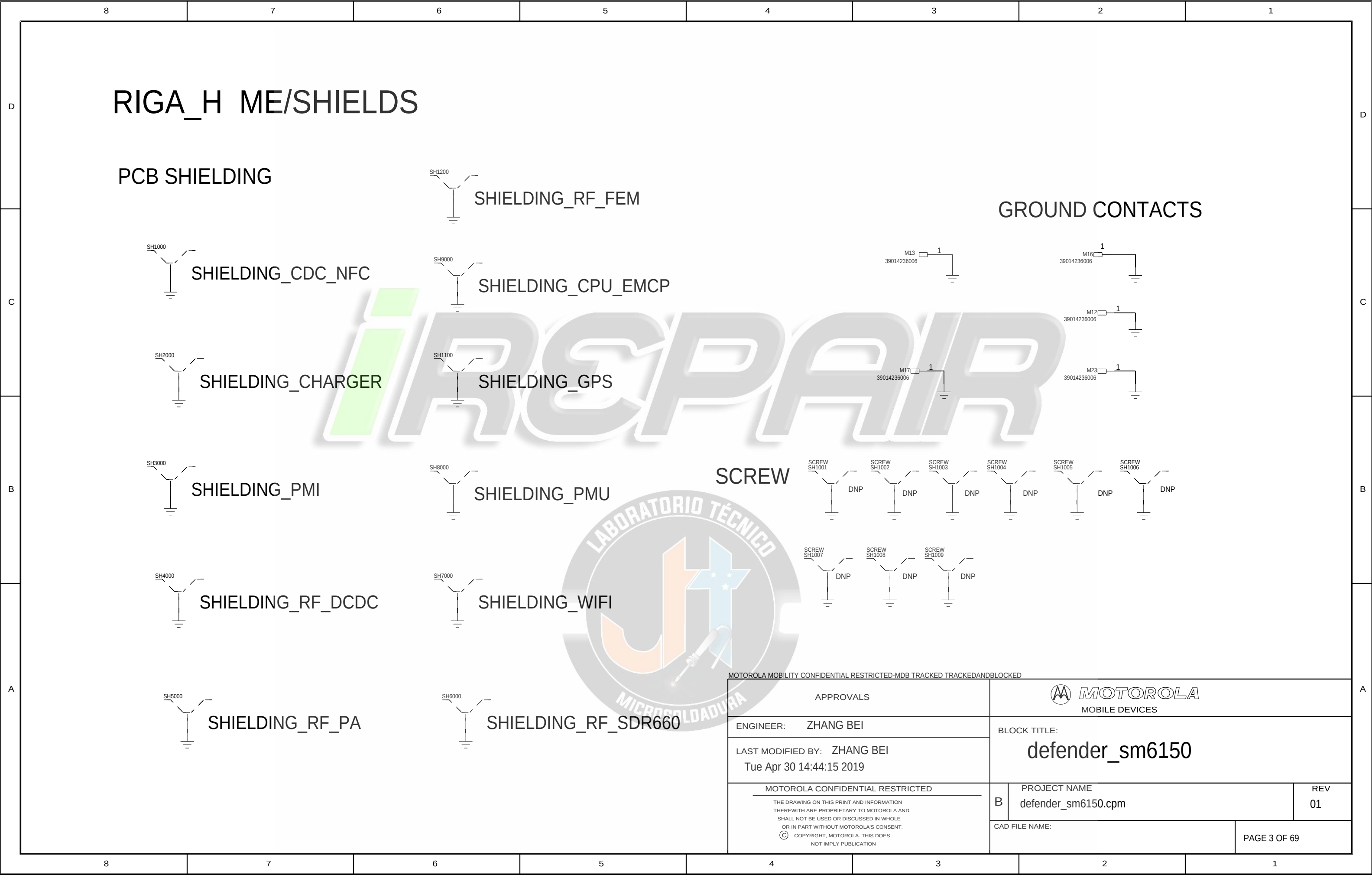
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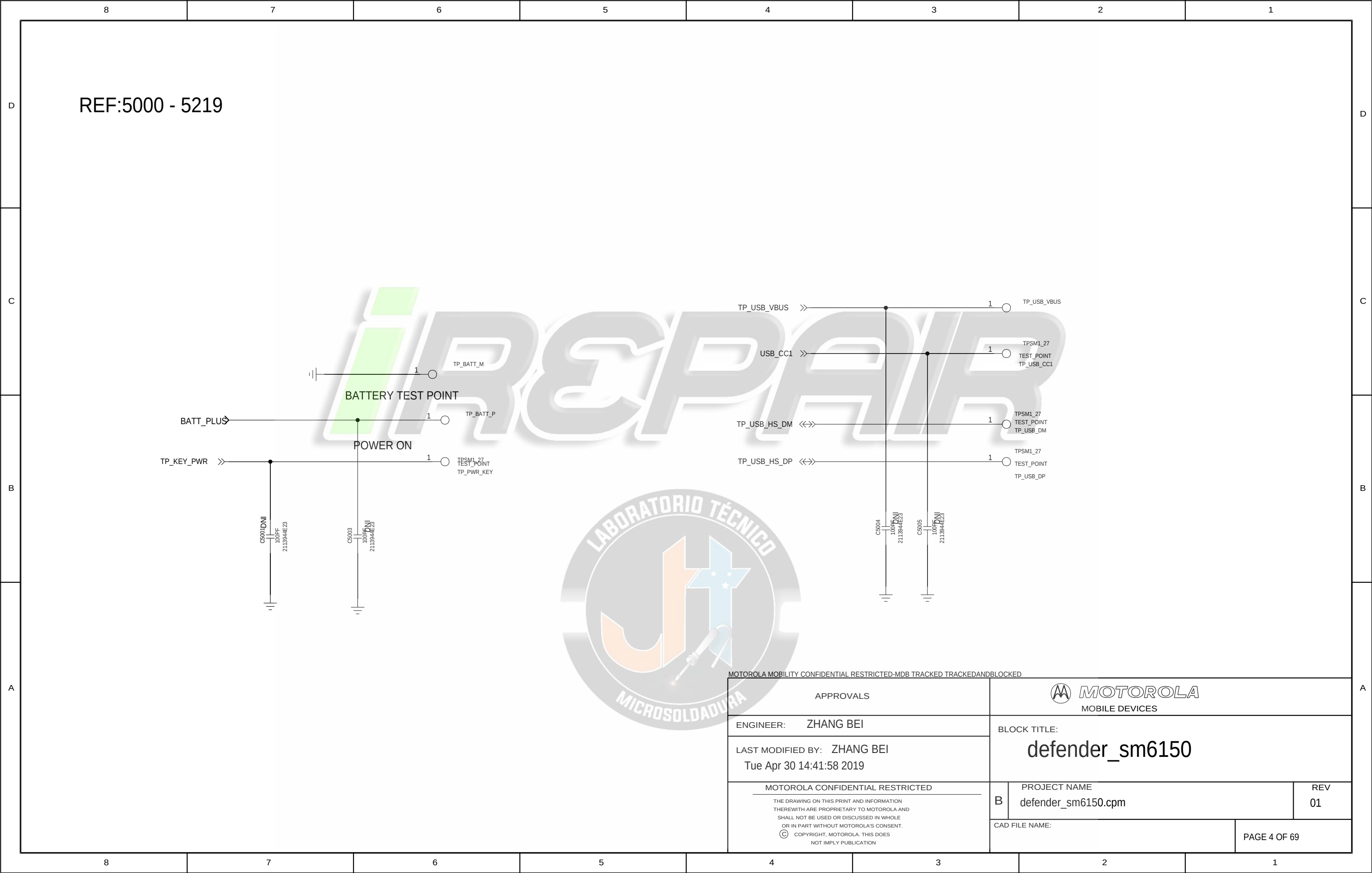
REVISIONS

ZONE	LTN	DESCRIPTION	DATE	APPR.
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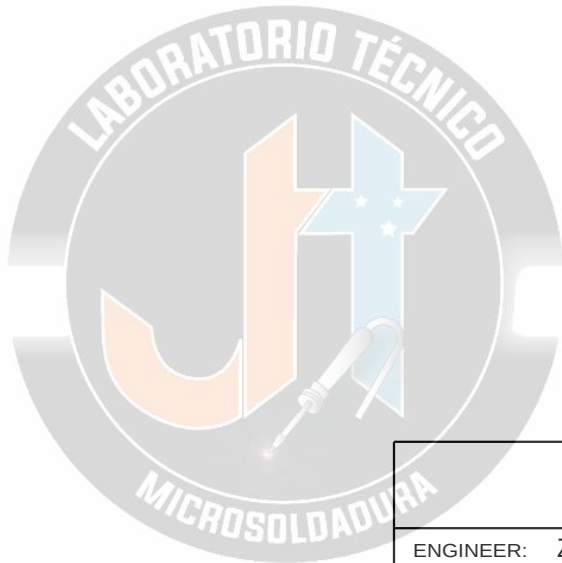
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MOTOROLA CONFIDENTIAL RESTRICTED		REV	01
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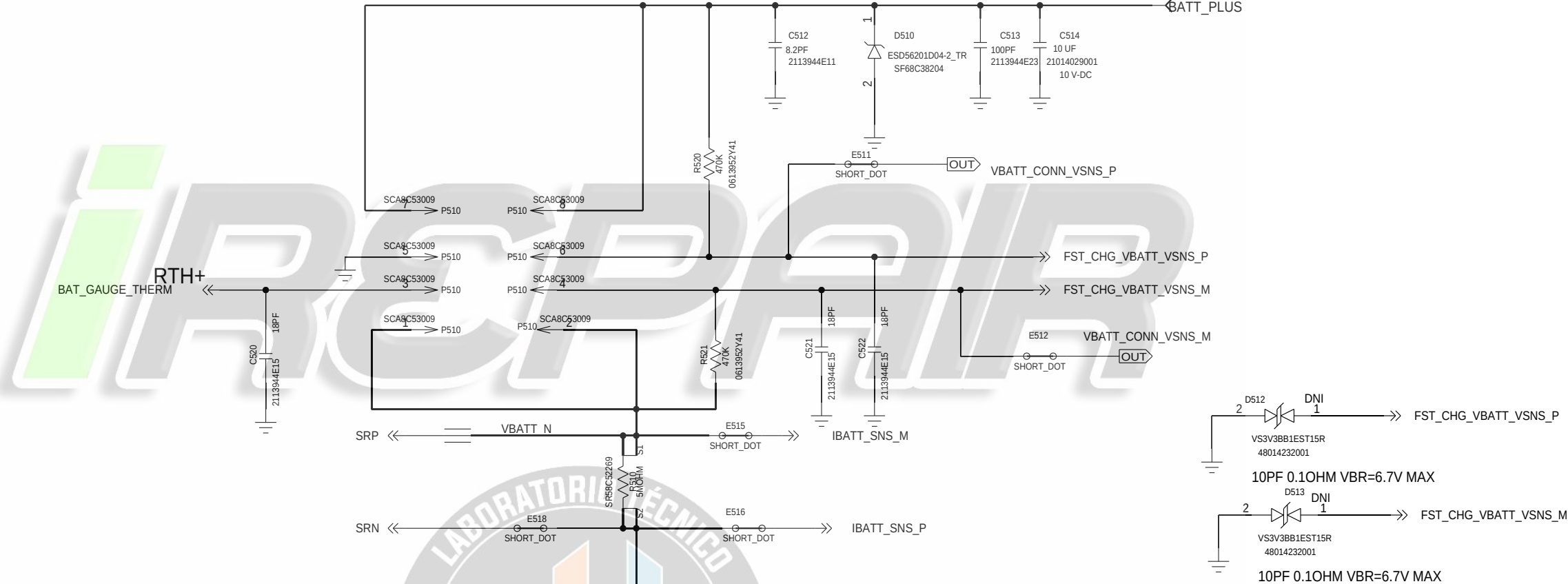
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PWR: USB-PD							
D							D
C							C
B							B
A							A
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iREPAIR



APPROVALS		MOTOROLA MOBILE DEVICES	
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		REV 01	
		CAD FILE NAME:	
		PAGE 5 OF 69	

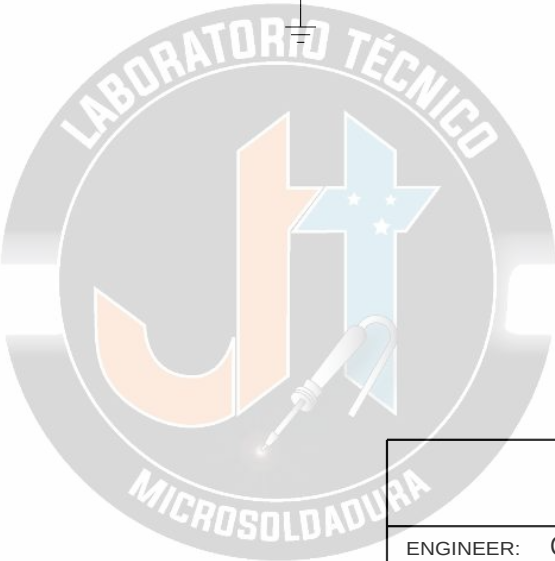
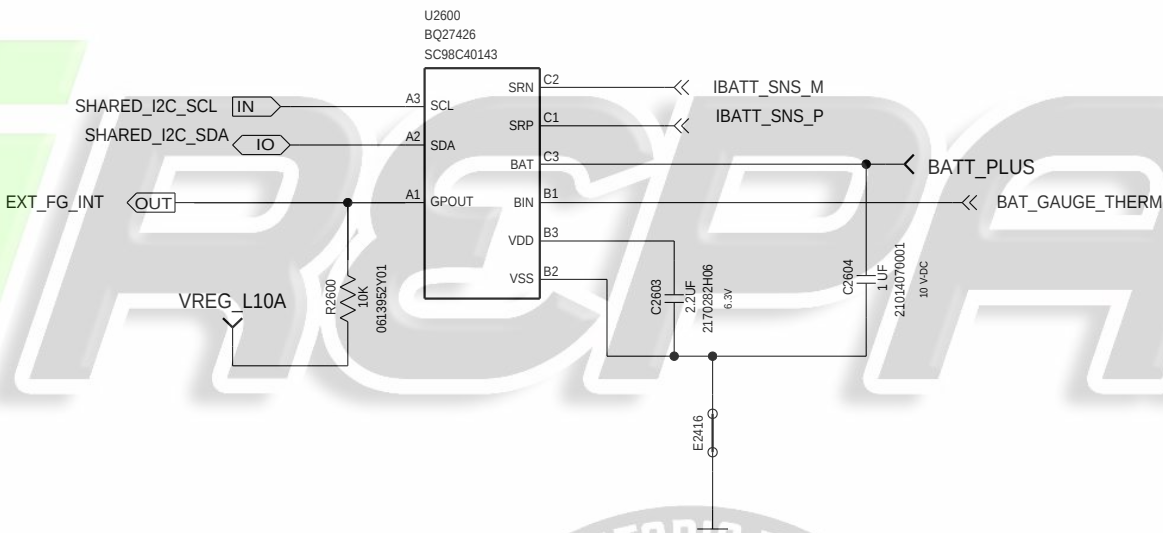
PWR: BATT
REF:510 - 550



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LAST MODIFIED BY: ZHANG BEI Tue Apr 30 14:41:59 2019		power	
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		REV 01 PAGE 6 OF 69	

PWR: FUEL GAUGE

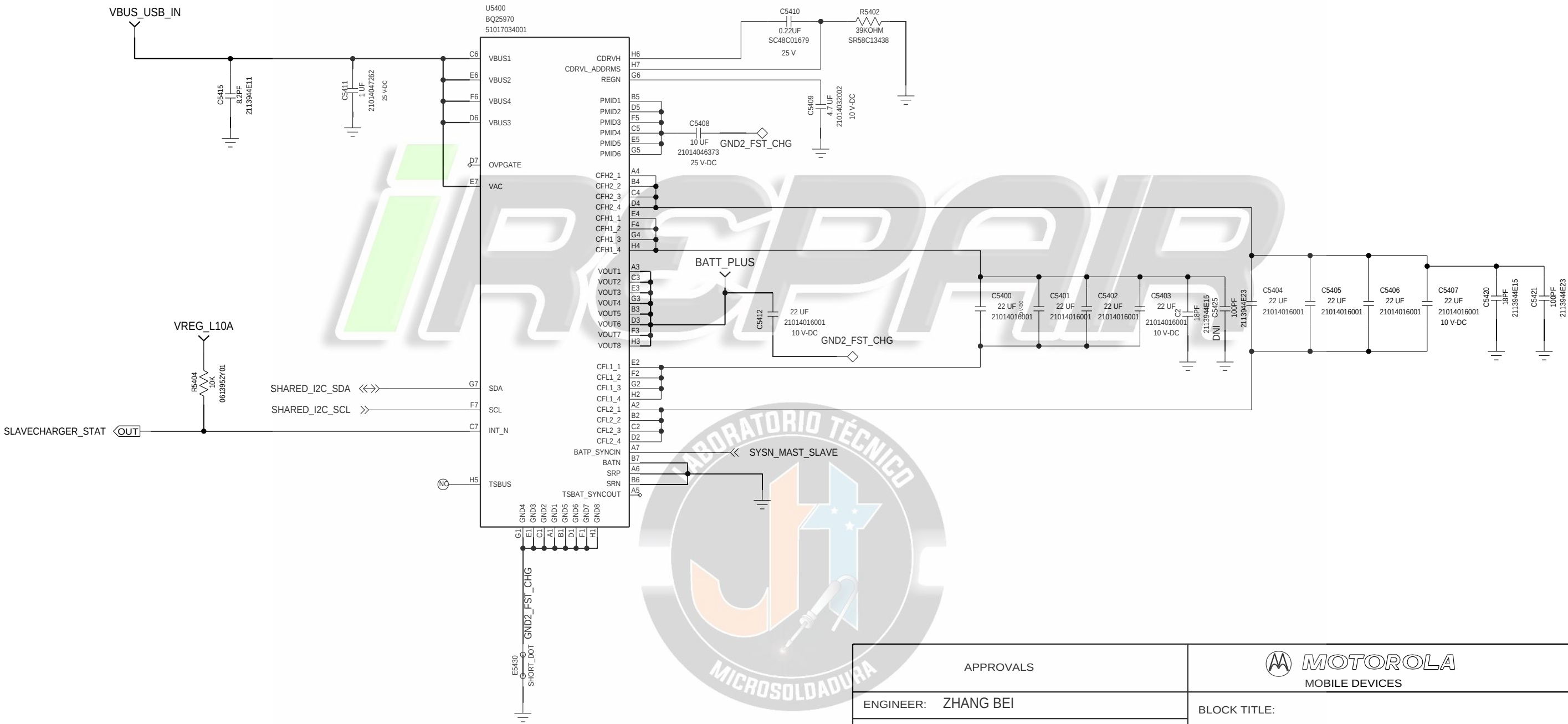
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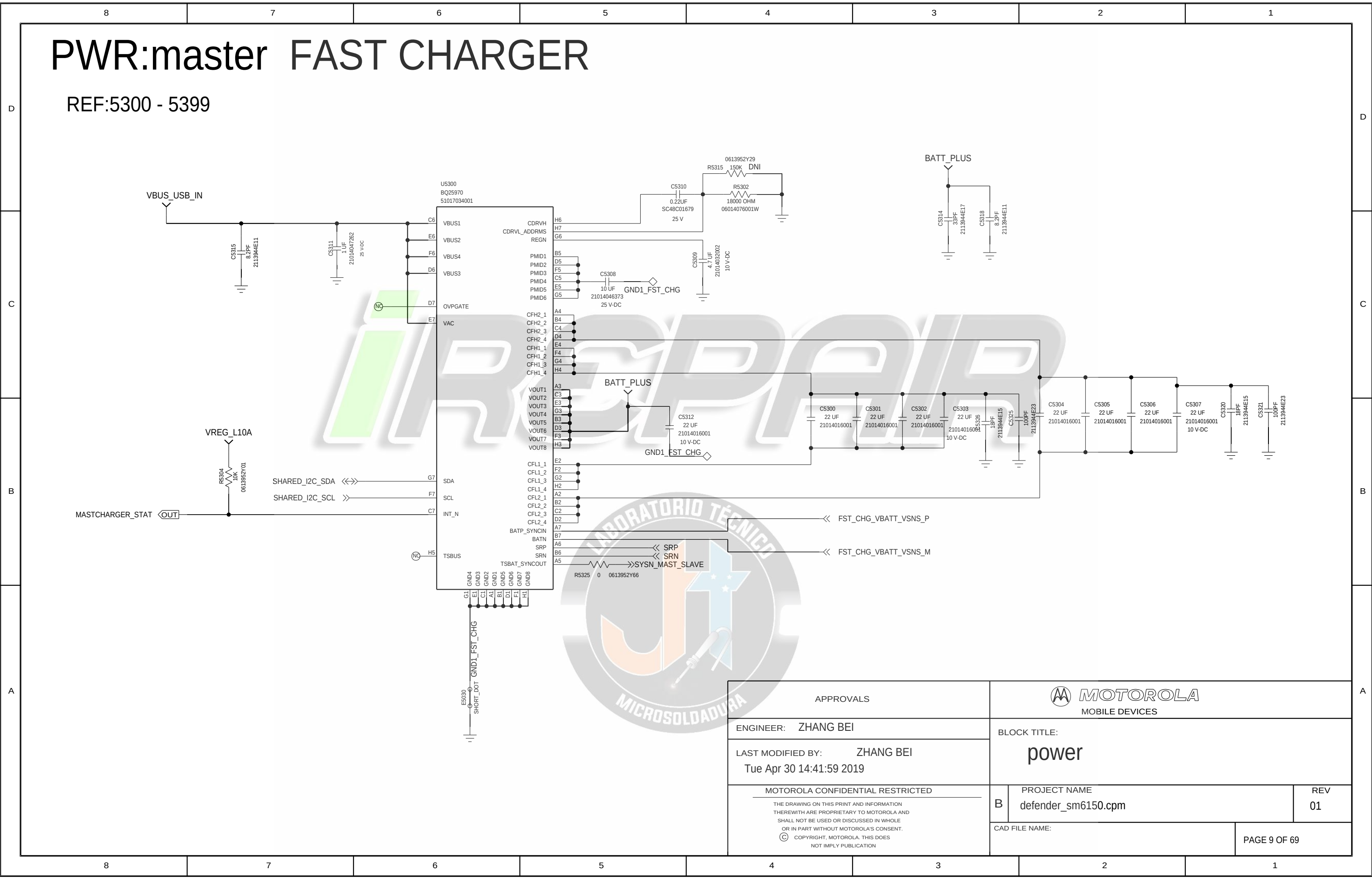
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		CAD FILE NAME:	
		PAGE 7 OF 69	

PWR:SLAVE FAST CHARGER

REF:5400 - 5499



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: ZHANG BEI		BLOCK TITLE:	
LAST MODIFIED BY: ZHANG BEI Tue Apr 30 14:41:59 2019		power	
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		REV 01	
		PAGE 8 OF 69	

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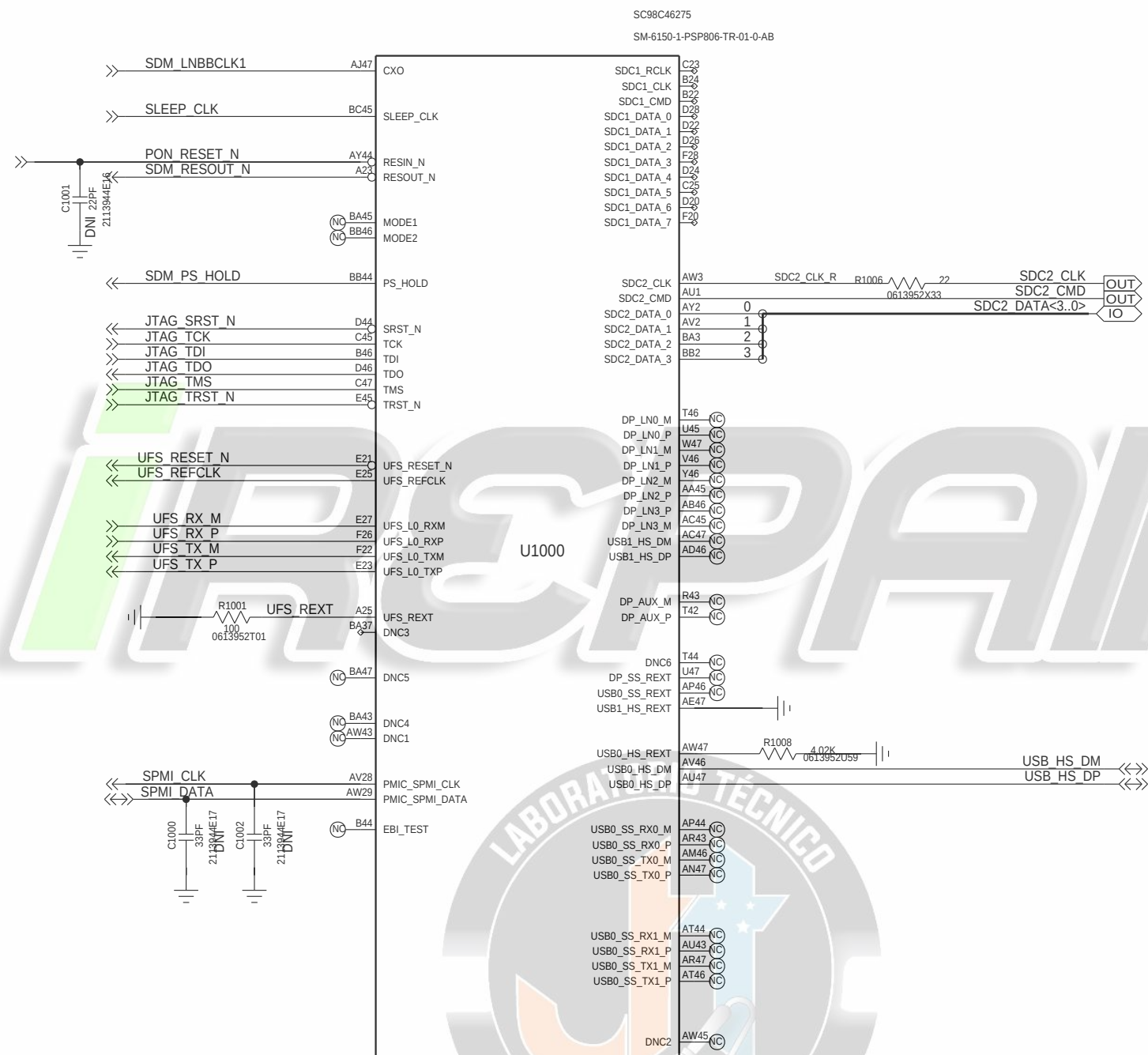
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PWR:master FAST CHARGER

REF:5300 - 5399

SM6150: CONTROL

REF 1000-1099

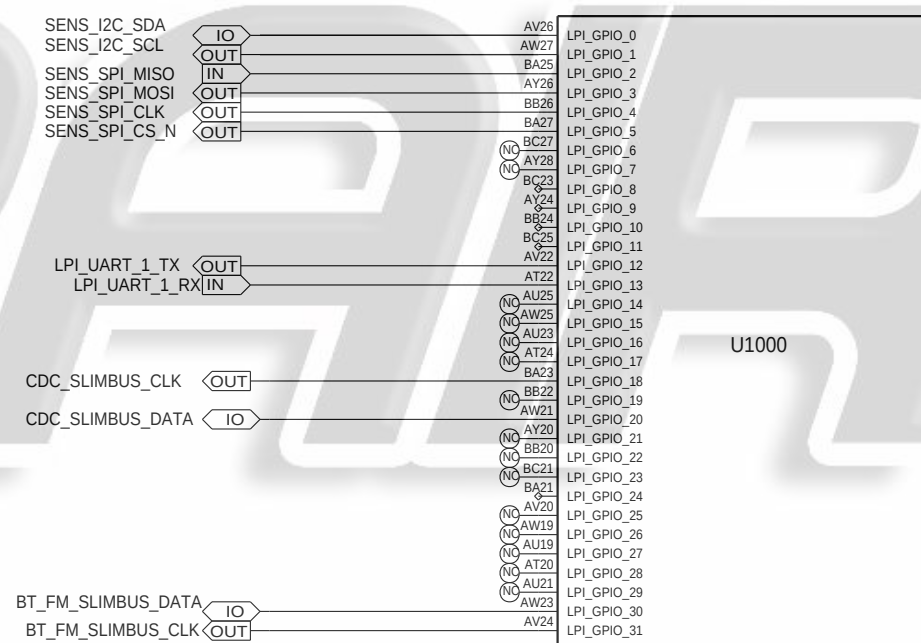
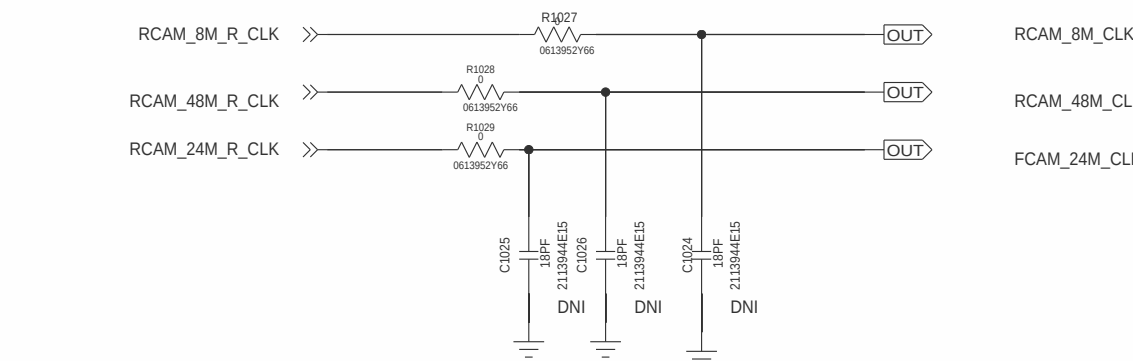
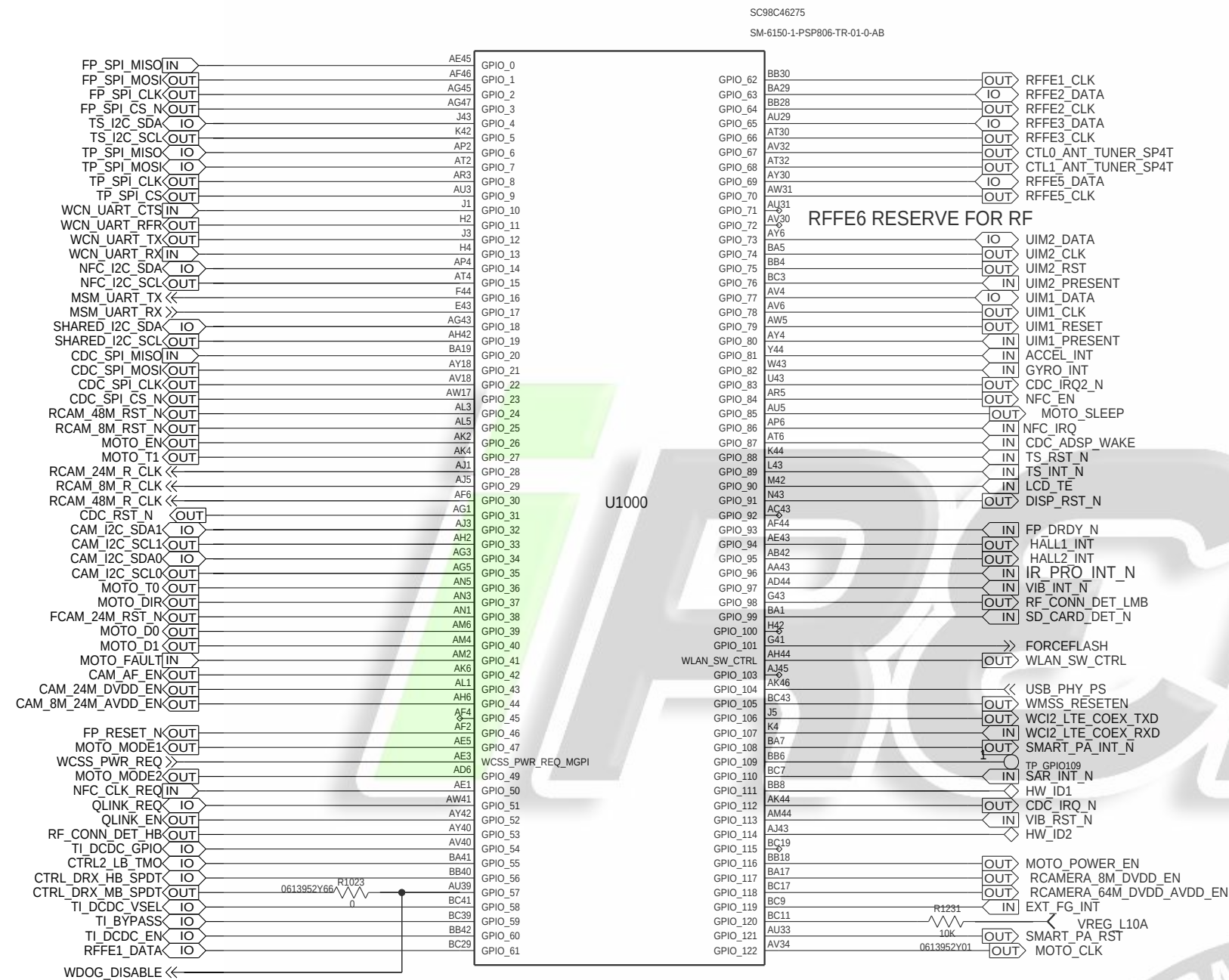


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ENGINEER: MOTOROLA		BLOCK TITLE:	
LAST MODIFIED BY: MOTOROLA Tue Apr 30 14:42:01 2019		ap_pmic	
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		REV 01	PAGE 10 OF 69

SM6150: GPIOs

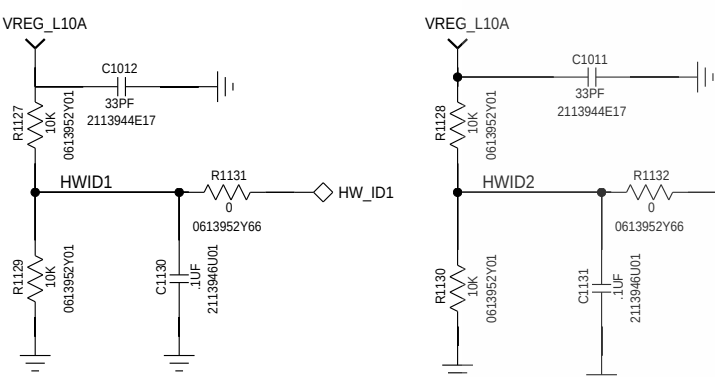
REVISIONS

ZONE	LTR	DESCRIPTION	DATE	APPR.

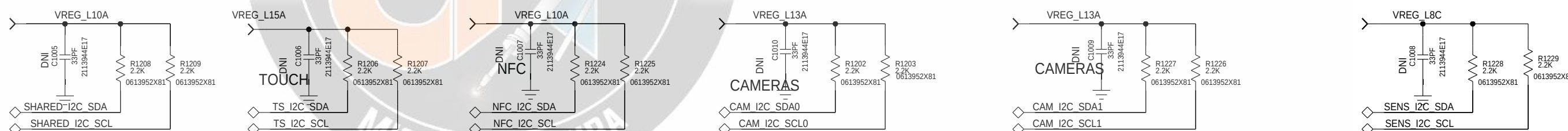


MOT BOARD ID

FOR RUNTIME REVISION INFORMATION, REFER TO:
<https://sites.google.com/a/motorola.com/runtime-product-ids>



I2C BUS PULL-UPS



APPROVALS

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MOBILE DEVICES

BLOCK TITLE:

ap_pmic

PROJECT NAME
defender_sm6150.cpm

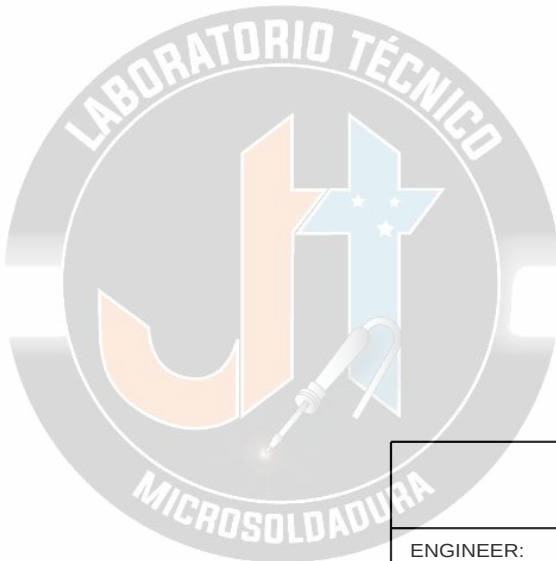
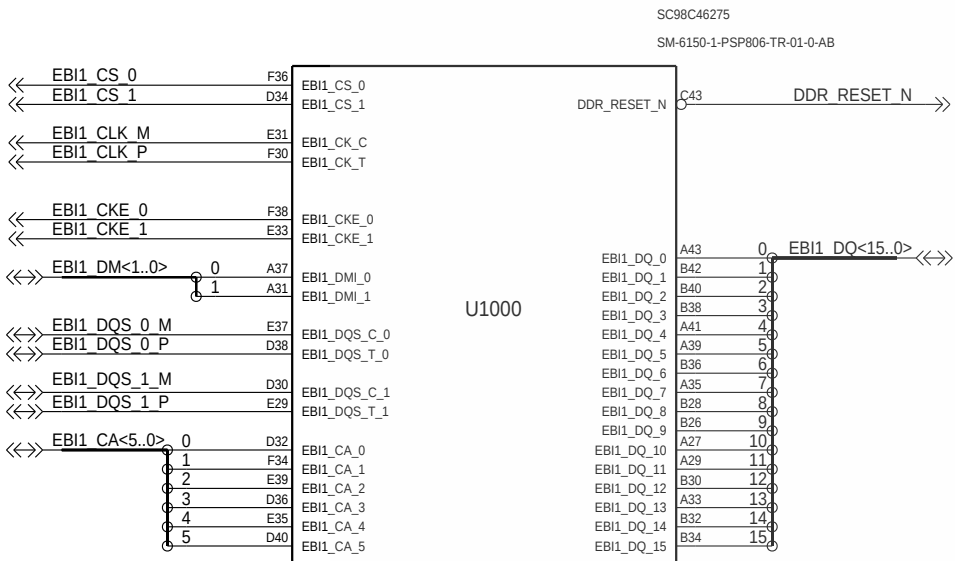
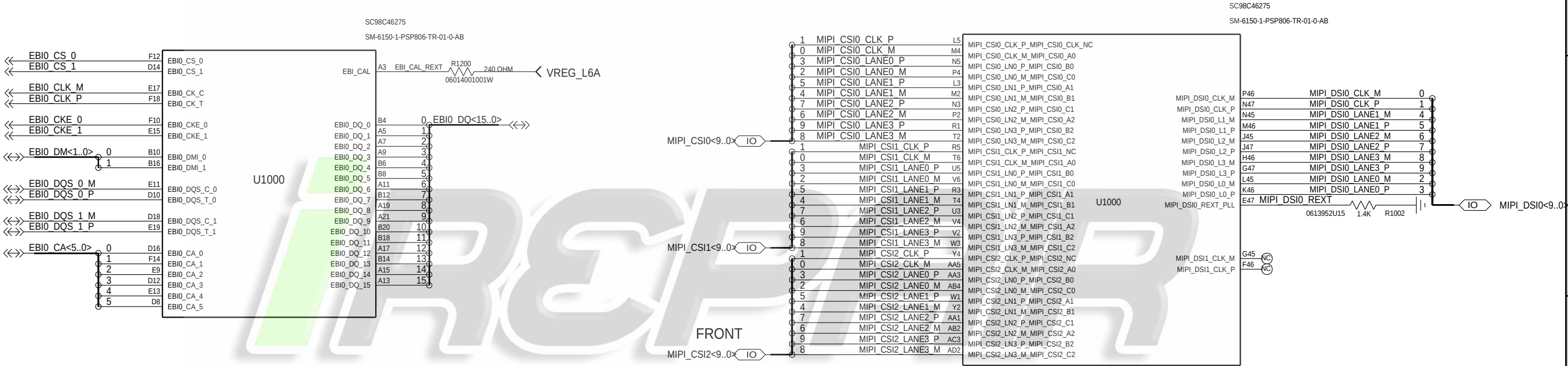
CAD FILE NAME:

REV
01

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SM6150: MIPI/DSI/CSI/EBI0

REF 1200-1299



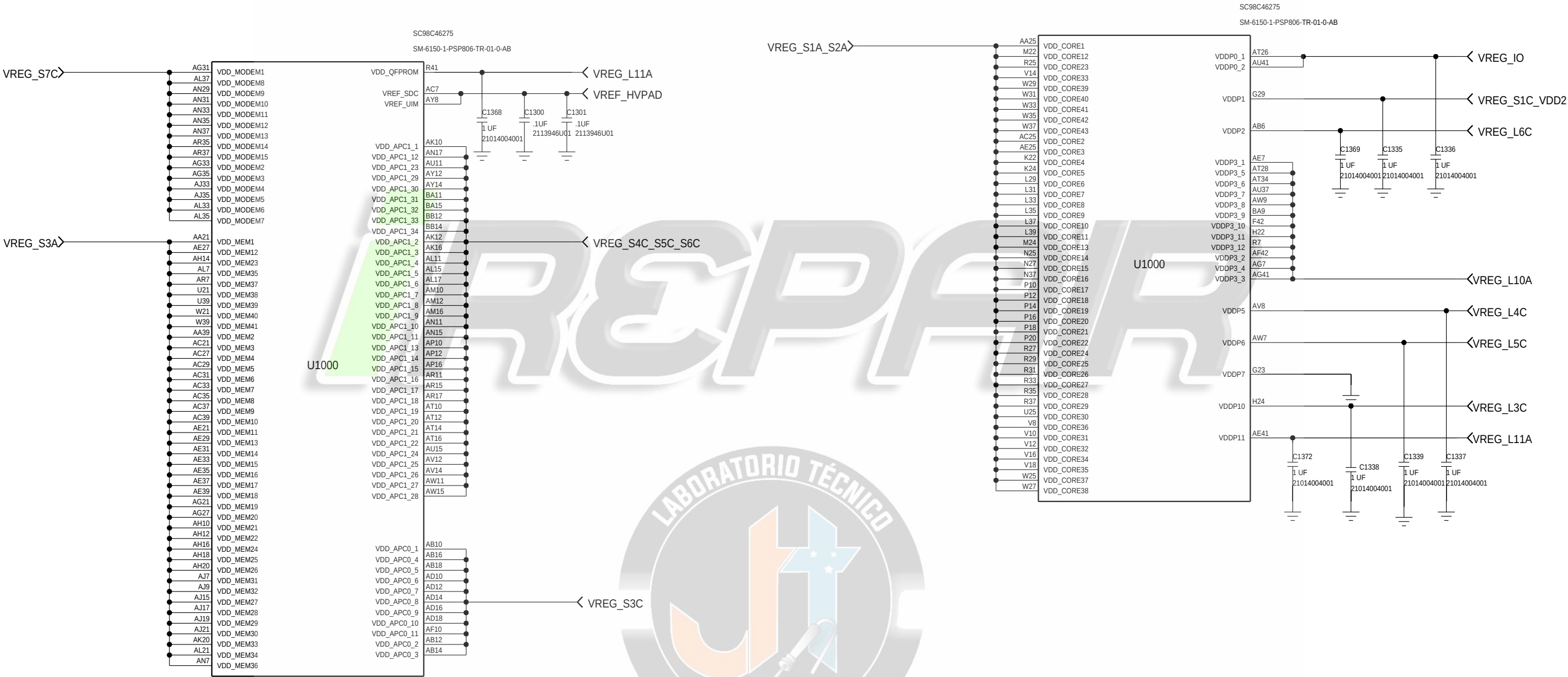
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		PAGE 12 OF 69	

D

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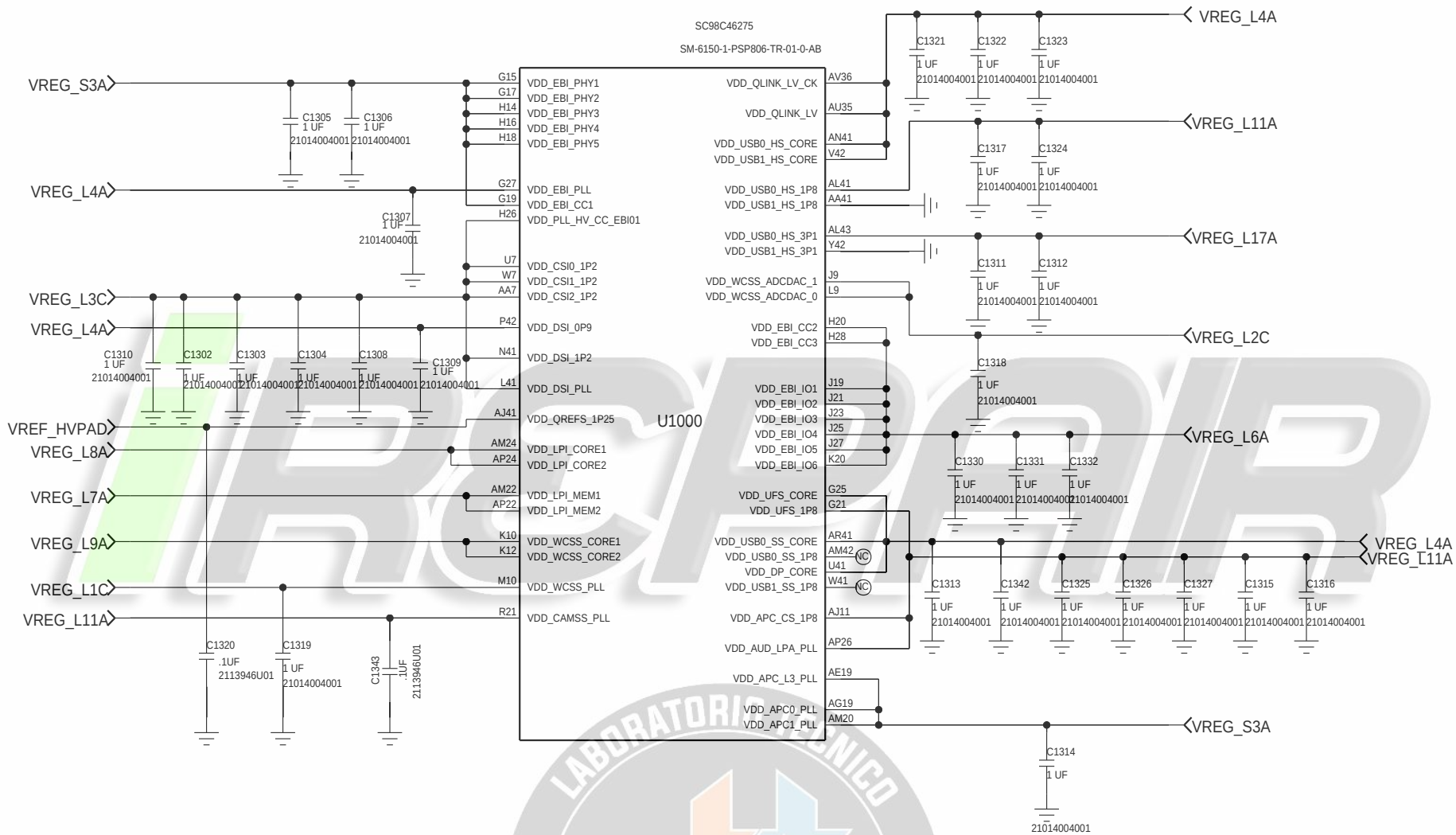
SM6150: PWR1/PWR2

REF 1300-1399



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		PAGE 14 OF 69	

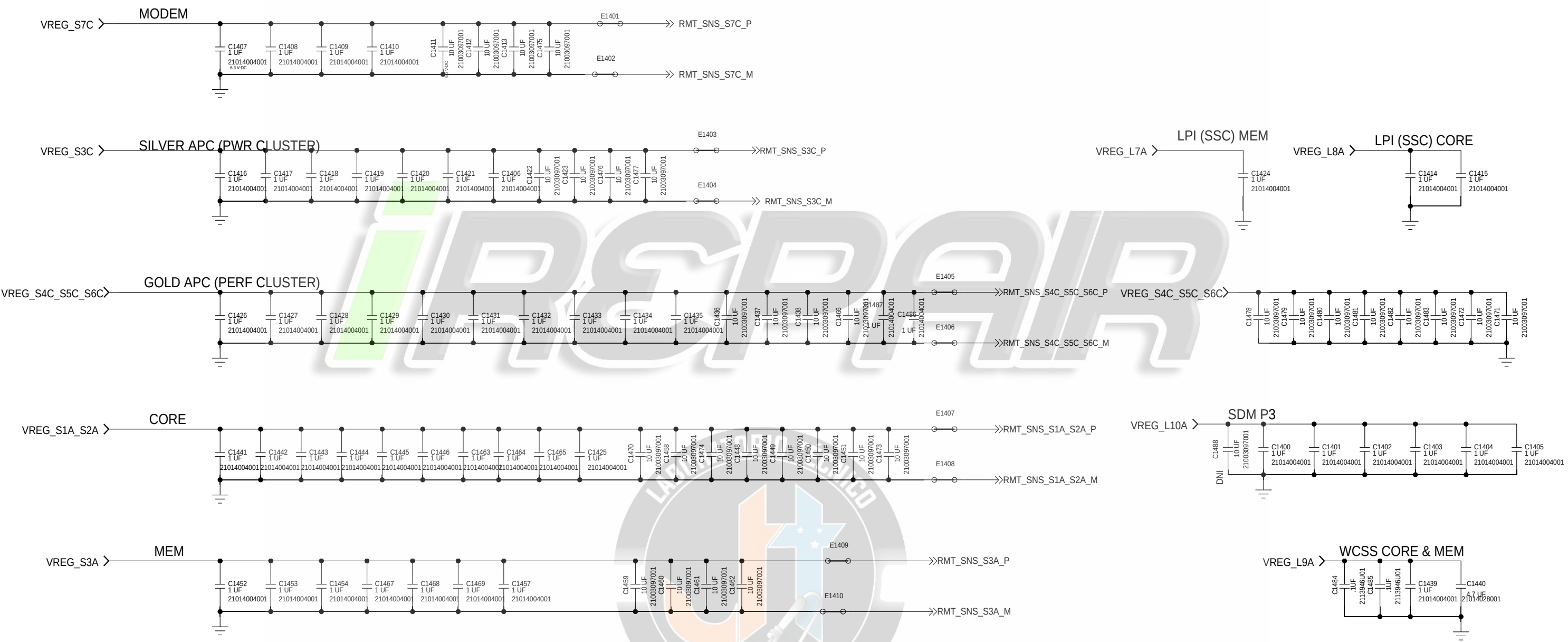
SM6150: PWR3
REF 1300-1399



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		CAD FILE NAME:	
		PAGE 15 OF 69	

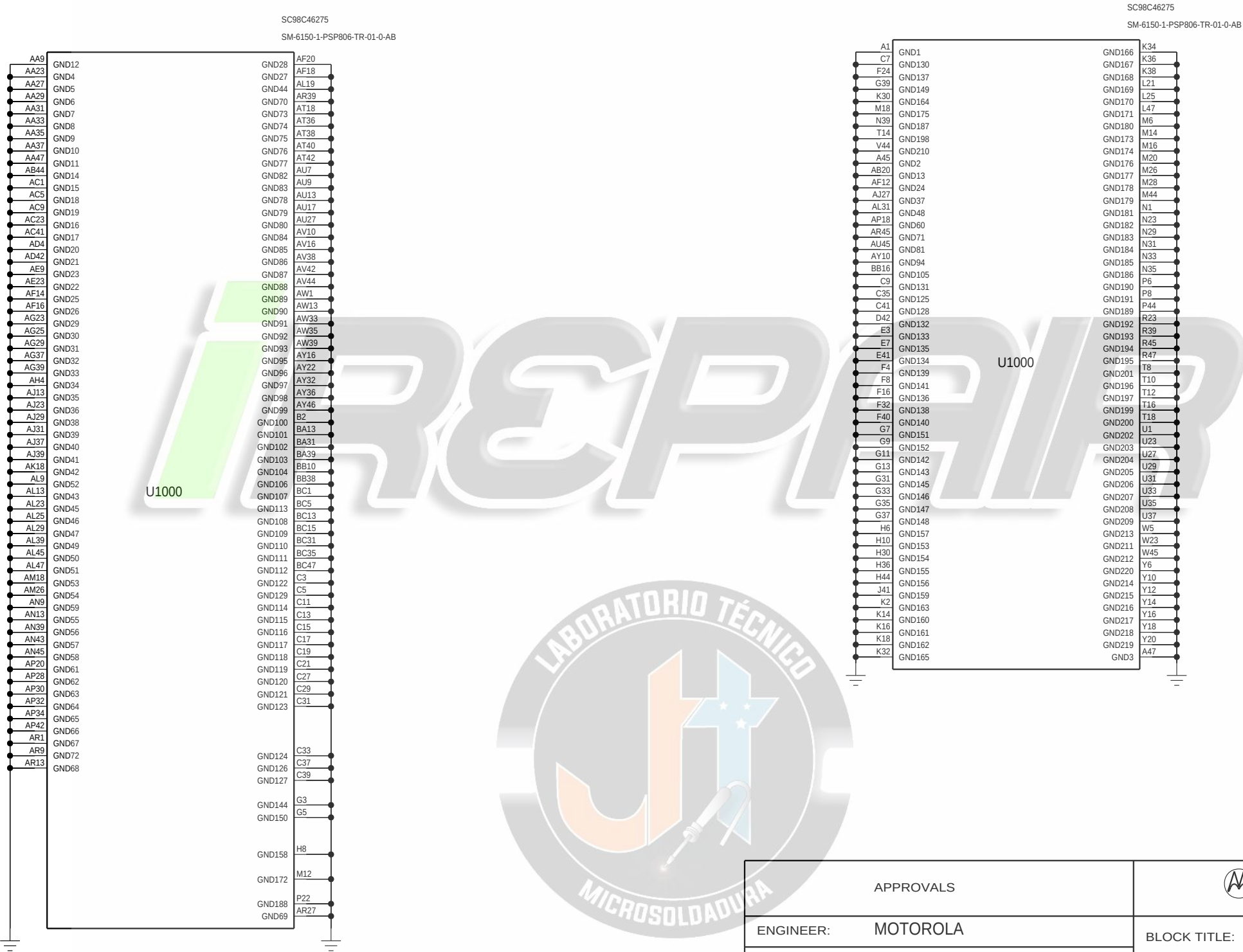
SM6150: DECAPS

REF 1400-1499



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LAST MODIFIED BY: MOTOROLA Tue Apr 30 14:42:03 2019		ap_pmic	
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		CAD FILE NAME:	
		PAGE 16 OF 69	

SM6150: GND

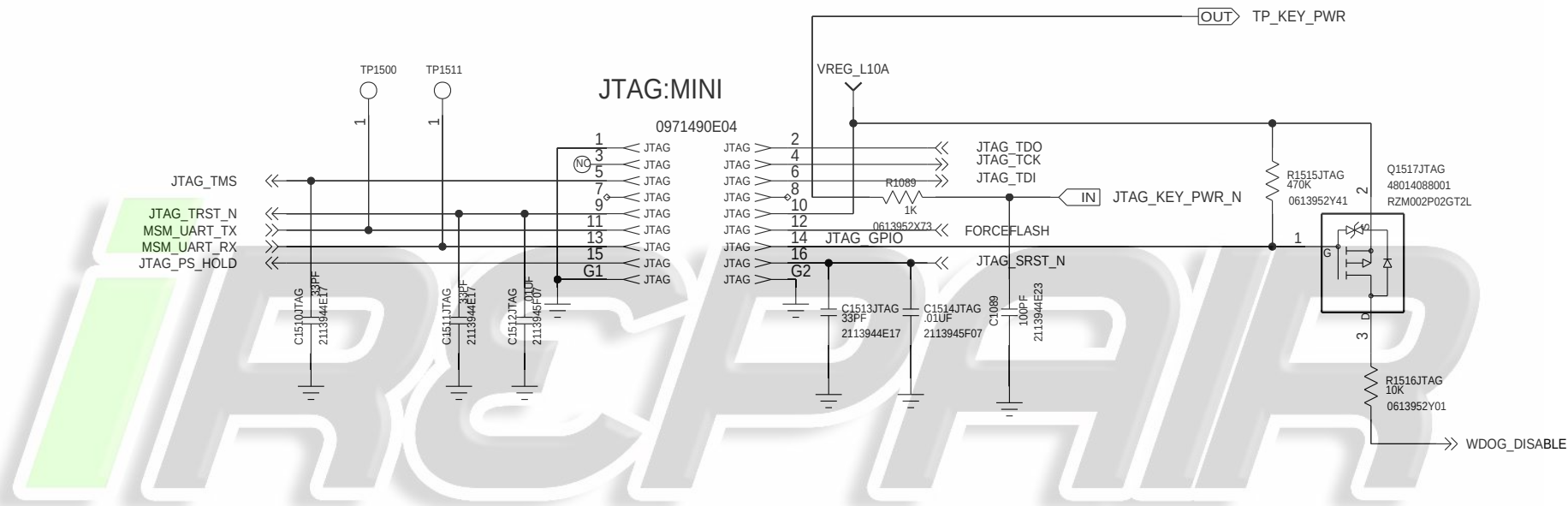


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ENGINEER: MOTOROLA		BLOCK TITLE:	
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		B defender_sm6150.cpm	REV 01
		CAD FILE NAME:	
		PAGE 17 OF 69	

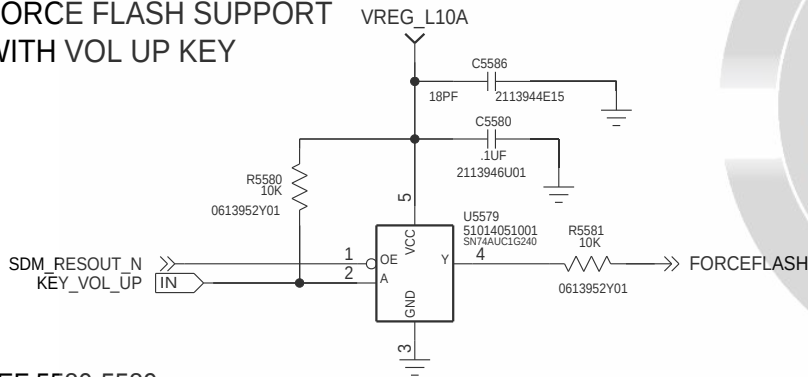
SM6150: DEBUG

REF 1000-1099

(DNI FOR PRODUCTION)



FORCE FLASH SUPPORT WITH VOL UP KEY



REF:5580-5589

APPROVALS

ENGINEER: MOTOROLA

LAST MODIFIED BY: MOTOROLA
Tue Apr 30 14:42:00 2019

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MOTOROLA
MOBILE DEVICES

BLOCK TITLE:

ap_pmic

PROJECT NAME
defender_sm6150.cpm

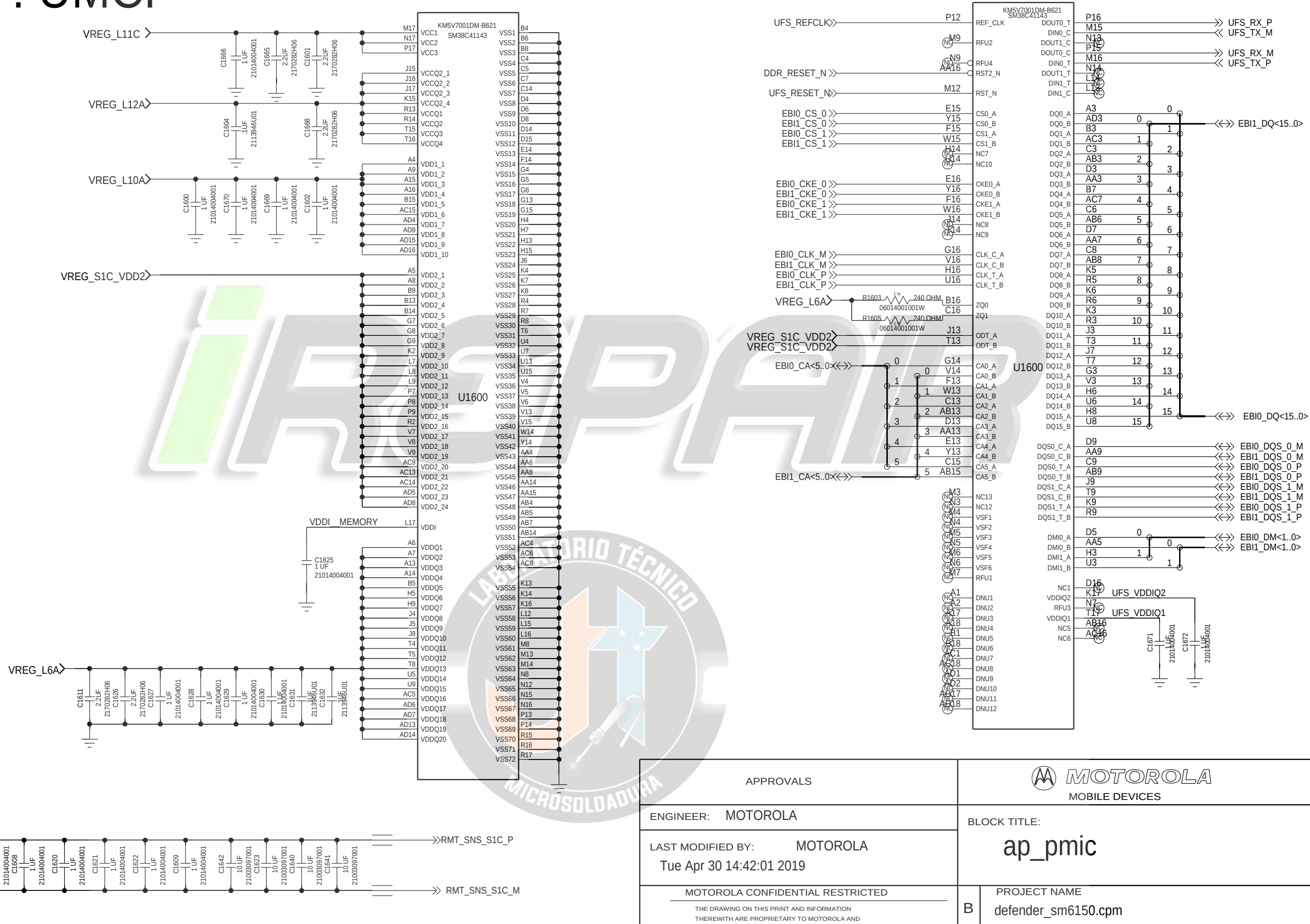
REV
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CAD FILE NAME:

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MEMORY: UMCP

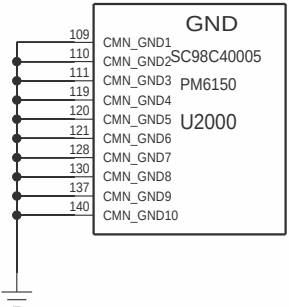
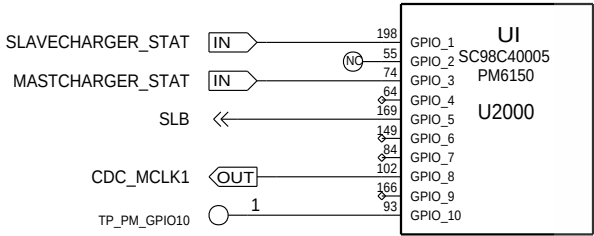
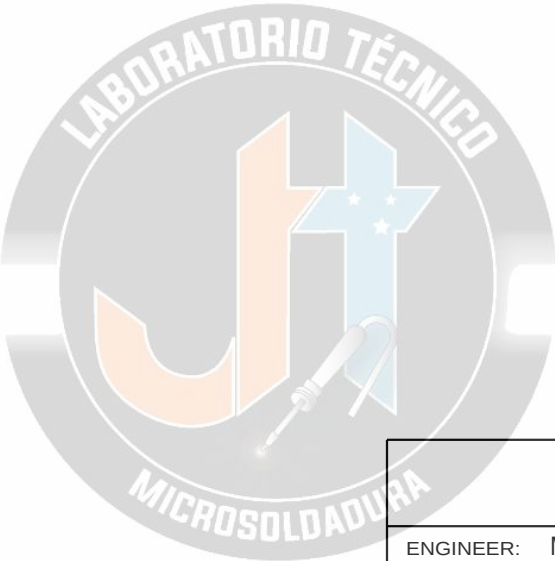
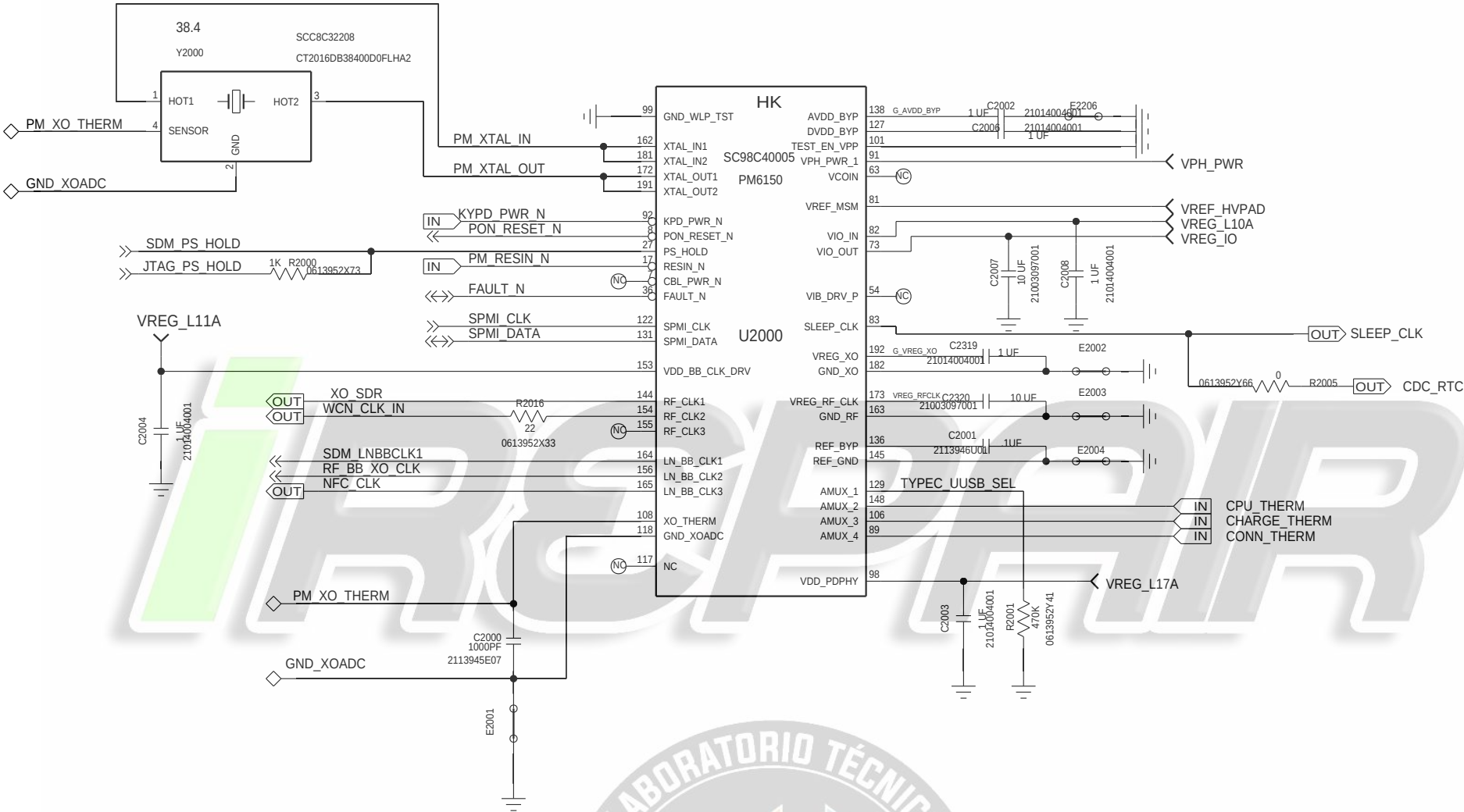
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		CAD FILE NAME:	
		PAGE 19 OF 69	

PM6150: CTRL/CLOCKS/GPIOS

REF: 2000-2099

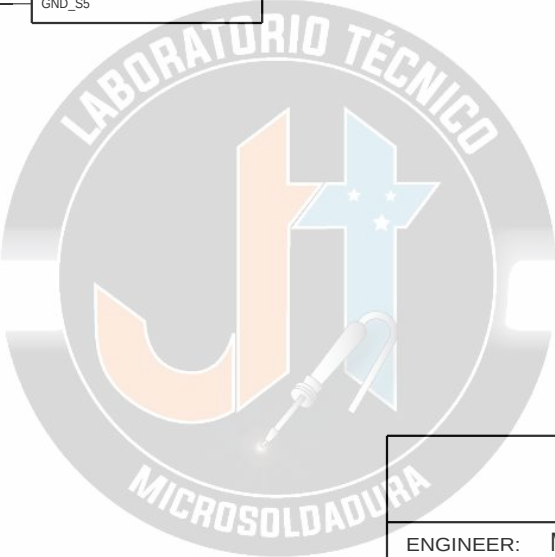
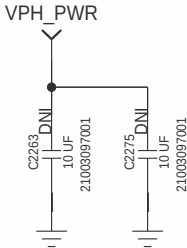
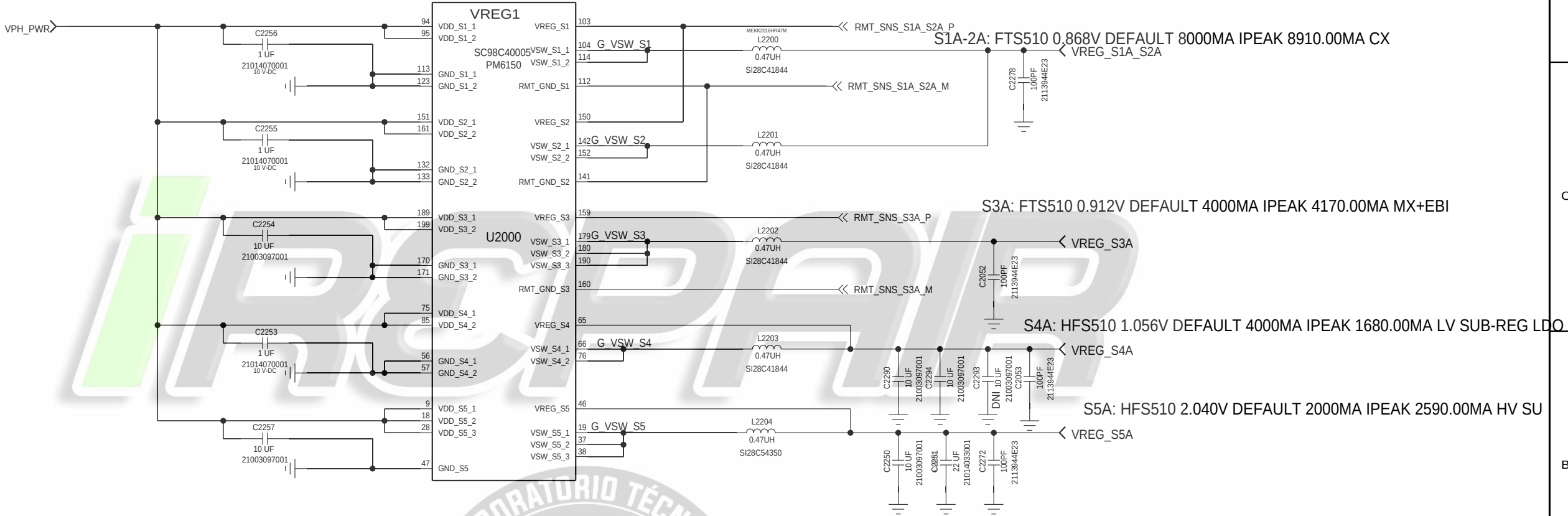


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LAST MODIFIED BY: MOTOROLA Tue Apr 30 14:42:00 2019		ap_pmic	
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		CAD FILE NAME:	
		PAGE 20 OF 69	

PM6150: SMPS

REF: 2200-2299

S1A-2A: FTS510 0.868V DEFAULT 8000MA IPEAK 8910.00MA CX



APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE:	
LAST MODIFIED BY: MOTOROLA Tue Apr 30 14:42:03 2019		ap_pmic	
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		CAD FILE NAME:	
		PAGE 21 OF 69	

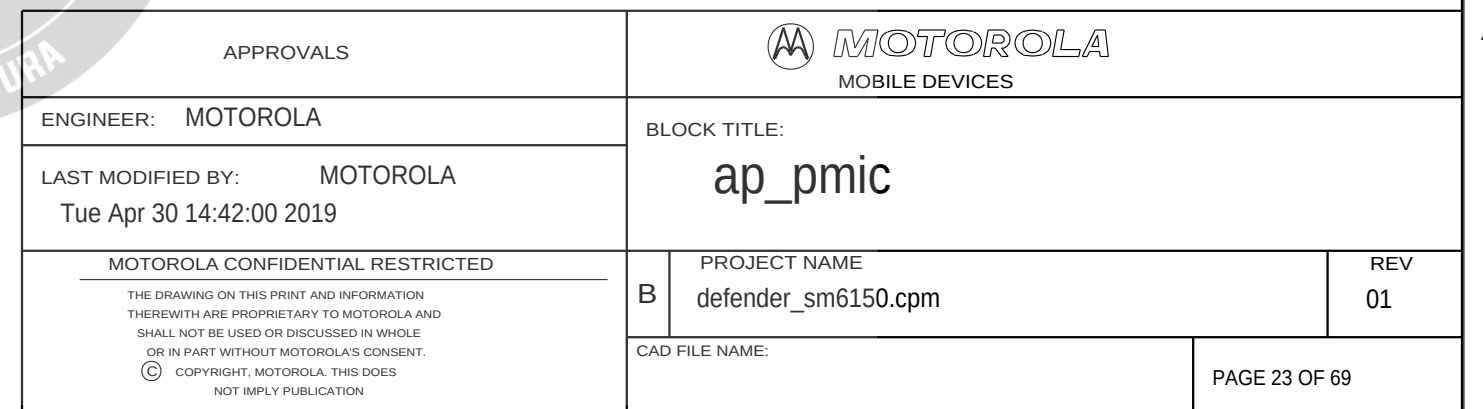
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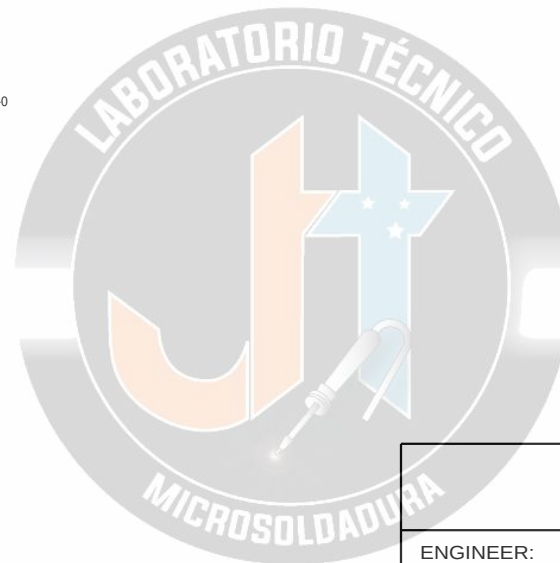
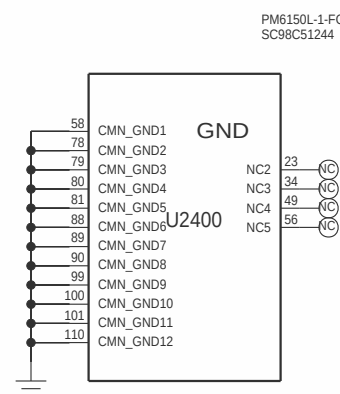
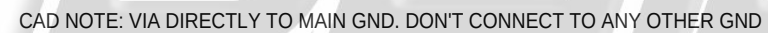
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REF: 2400-2419



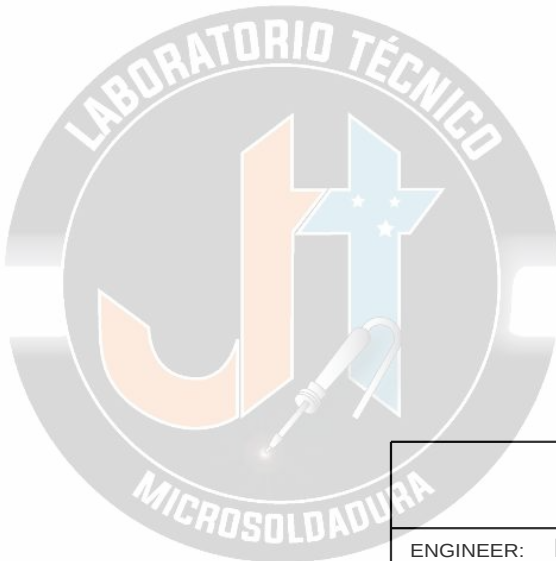
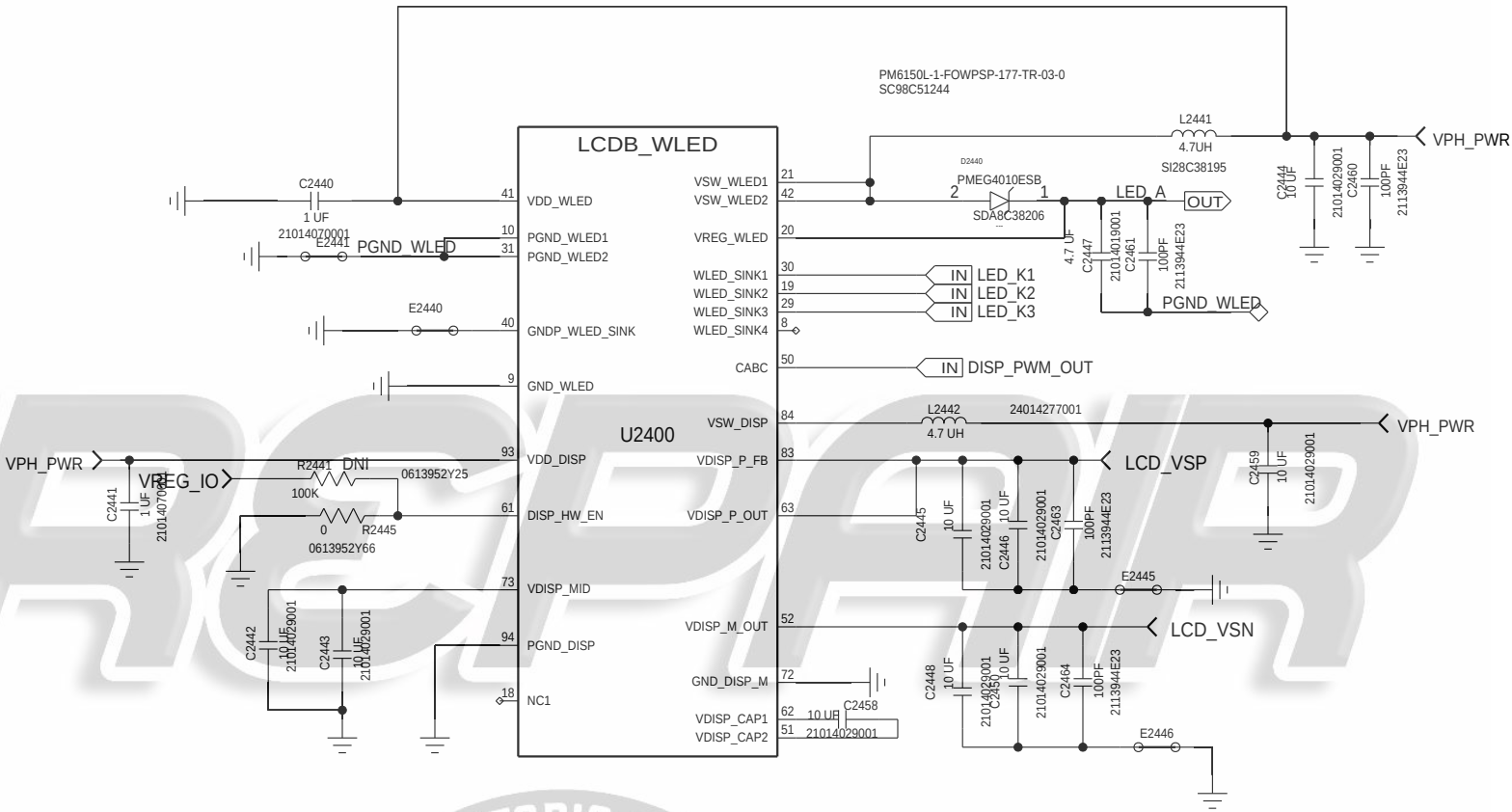
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APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA	BLOCK TITLE: ap_pmic		
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		CAD FILE NAME:	PAGE 24 OF 69

PM6150L: LCD_BIAS &&WLED

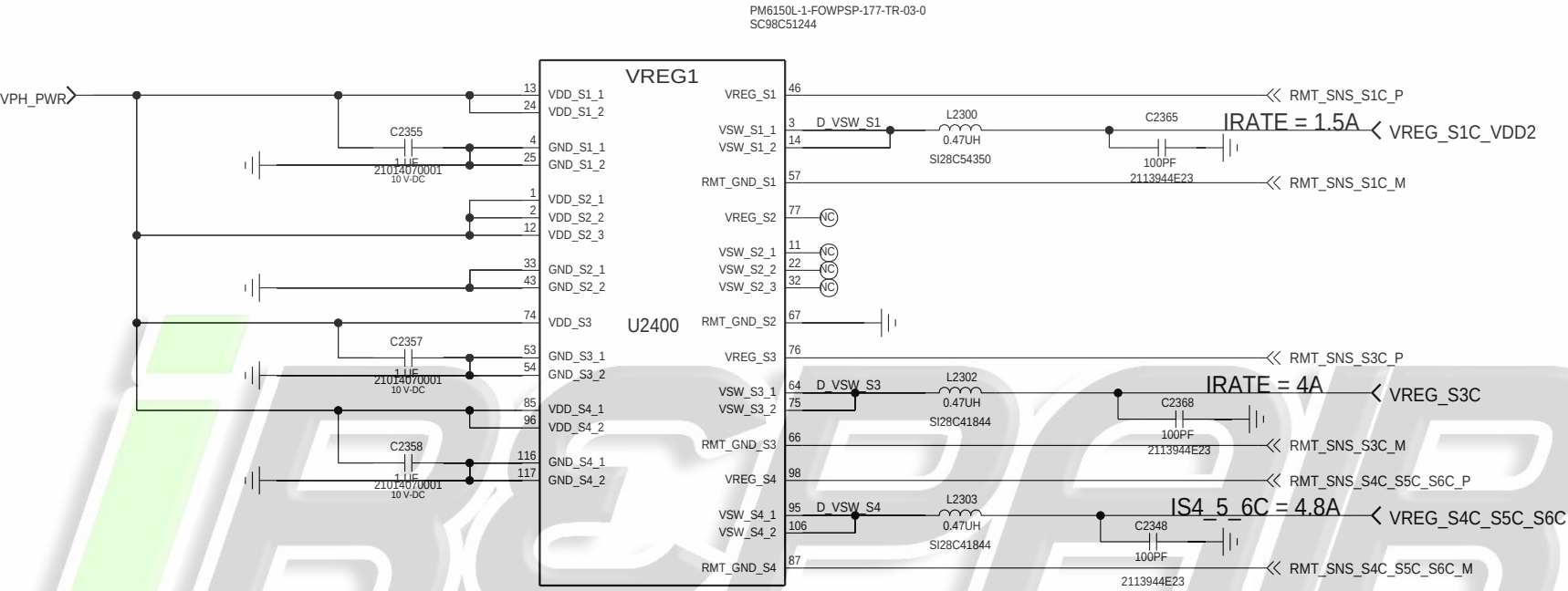
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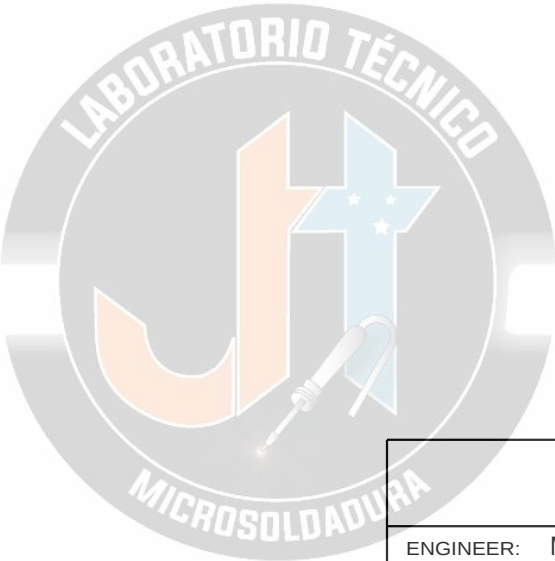
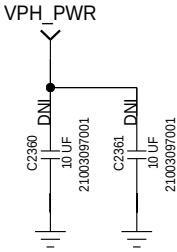
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		CAD FILE NAME:	
		PAGE 26 OF 69	

PM6150L: SMPS1

REF: 2300-2399



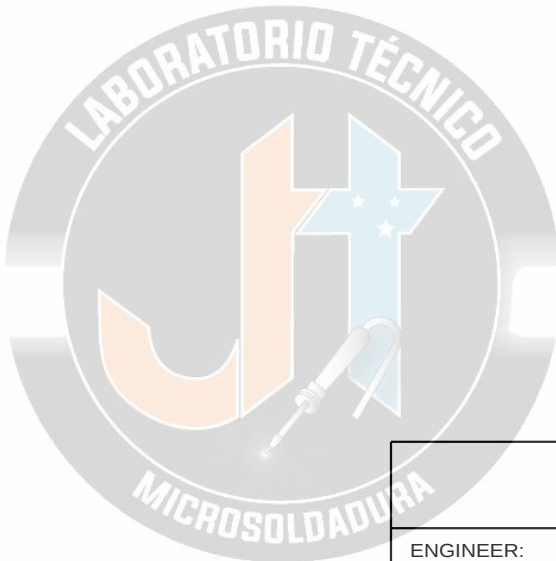
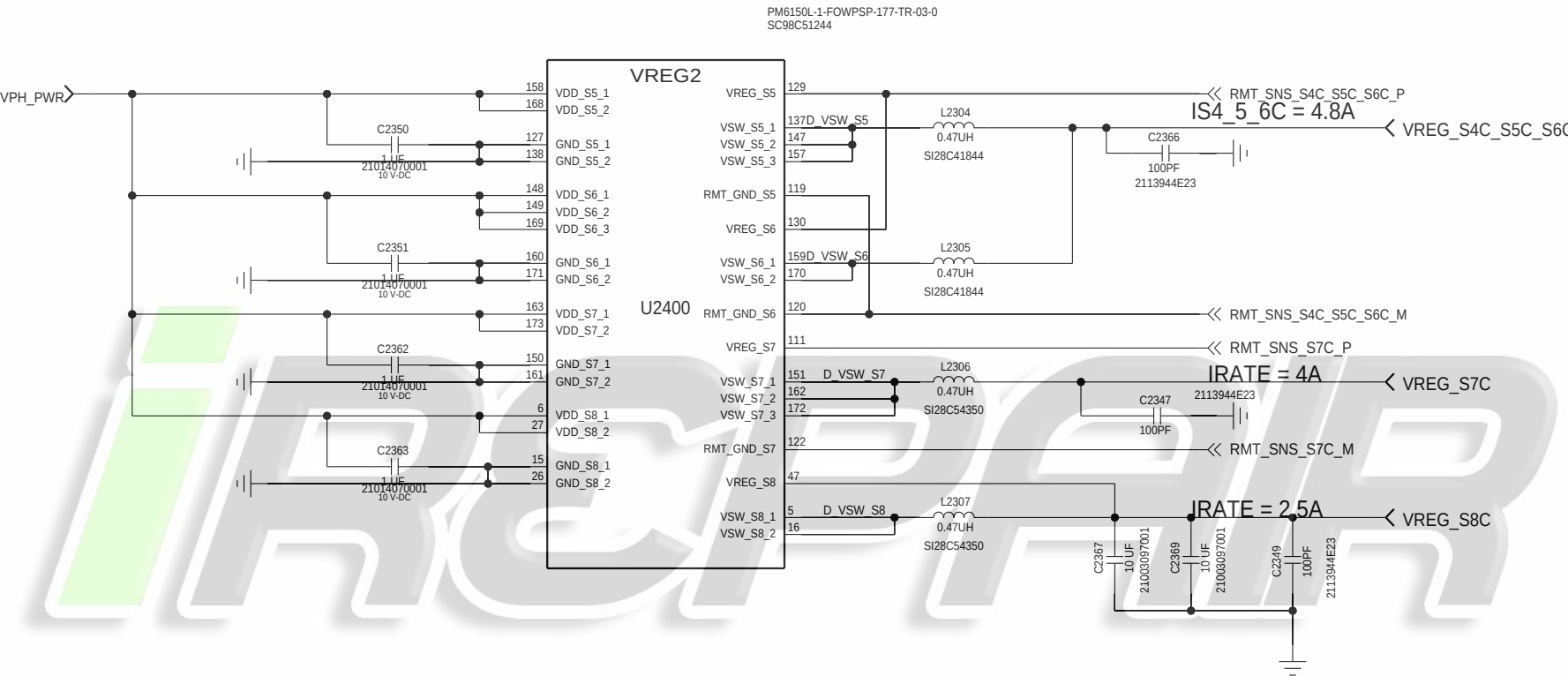
B_PLUS BULK_CAPS.DISTRIBUTE THESE ALL OVER U2400



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE:	
LAST MODIFIED BY: MOTOROLA Tue Apr 30 14:42:02 2019		ap_pmic	
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		REV 01	
		CAD FILE NAME:	
		PAGE 27 OF 69	

PM6150L: SMPS2

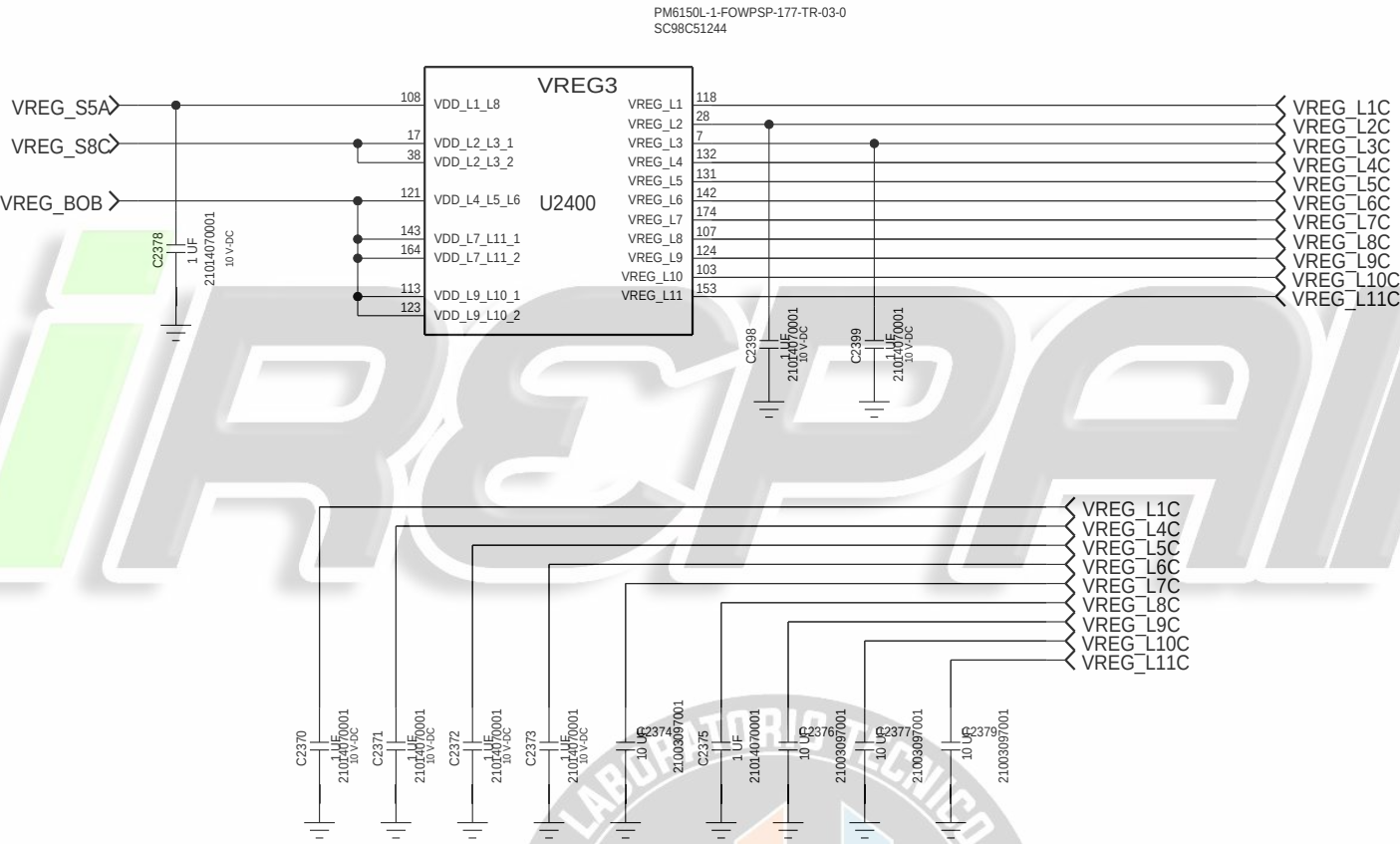
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APPROVALS		MOTOROLA MOBILE DEVICES	
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		CAD FILE NAME: PAGE 28 OF 69	

PM6150L: LDOS

REF: 2300-2399



PSEUDO CAPLESS LDOS
FOR CAPLESS LODS:IF DECAPS ON THE LOAD SIDE DO NOT ADD UP TO LDO SPEC, THEN INSTALL THE CAP CLOSE TO THE PMIC

Table 3-6 Linear/low-voltage regulator summary for PM6150A/PM6150L

Regulator	Circuit type	I _{rated} (mA)	Default on booting voltage (V) ¹	Nominal voltage (V)	Sleep state	Specified range (V)	Expected use
L1	LDO510 LV PMOS 150	150	-	1.8	Off	1.504 to 2	WCN 1.8 V XO
L2	LDO510 NMOS 600	600	-	1.304	Off	0.32 to 1.304	WCN 1.3 V RFA
L3	LDO510 NMOS 600	600	1.232	1.232	Off	0.32 to 1.304	1.23 V MSIPs
L4	LDO510 MV PMOS 150	150	-	1.808	Off	1.504 to 3.544	UIM1
L5	LDO510 MV PMOS 150	150	-	1.808	Off	1.504 to 3.544	UIM2
L6	LDO510 MV PMOS 150	150	2.96	2.96	Off	1.504 to 3.544	PX_2
L7	LDO510 MV PMOS 600	600	-	3.008	Off	1.504 to 3.544	ALS sensor 3 V, TSP 3 V
L8	LDO510 LV PMOS 150	150	-	1.8	Off	1.504 to 2	1.8 V environment sensors
L9	LDO510 MV PMOS 600	600 ²	2.96	2.96	Off	1.504 to 3.544	SD_MMC 3 V
L10	LDO510 MV PMOS 600	600	-	3.304	Off	1.504 to 3.544	WCN 3.3 V
L11	LDO510 MV PMOS 600	600	2.96	2.96	Off	1.504 to 3.544	UFS 2.96 V

1. Voltages shown in the Default on booting voltage (V) column indicate that the regulator is enabled as part of the hardware default-on sequence.
2. LDO9 is tested and used beyond its I_{rated} limit for SD3.0 support.

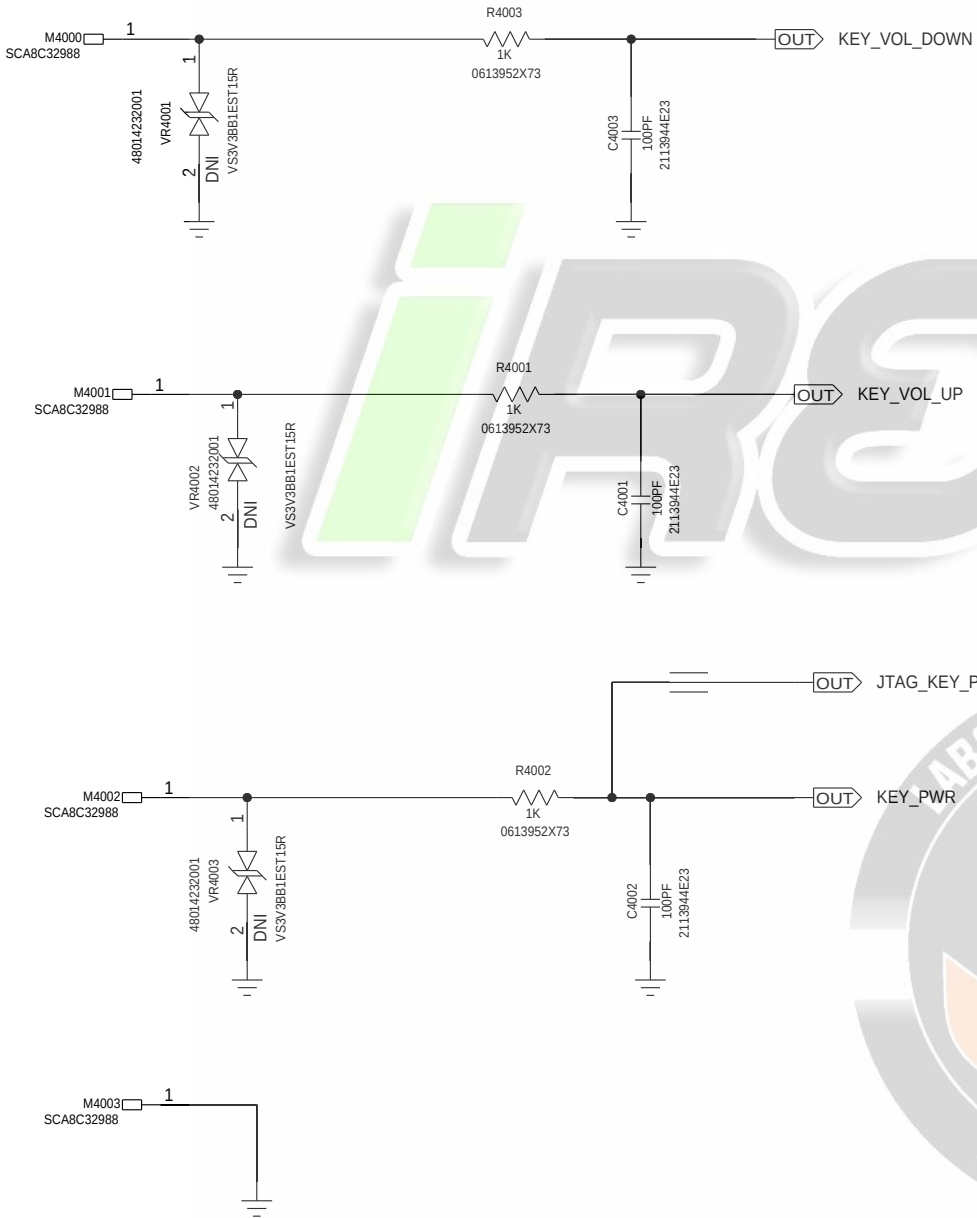
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		REV 01	
		PAGE 29 OF 69	

UI: SIDE KEY

REF:4000-4199

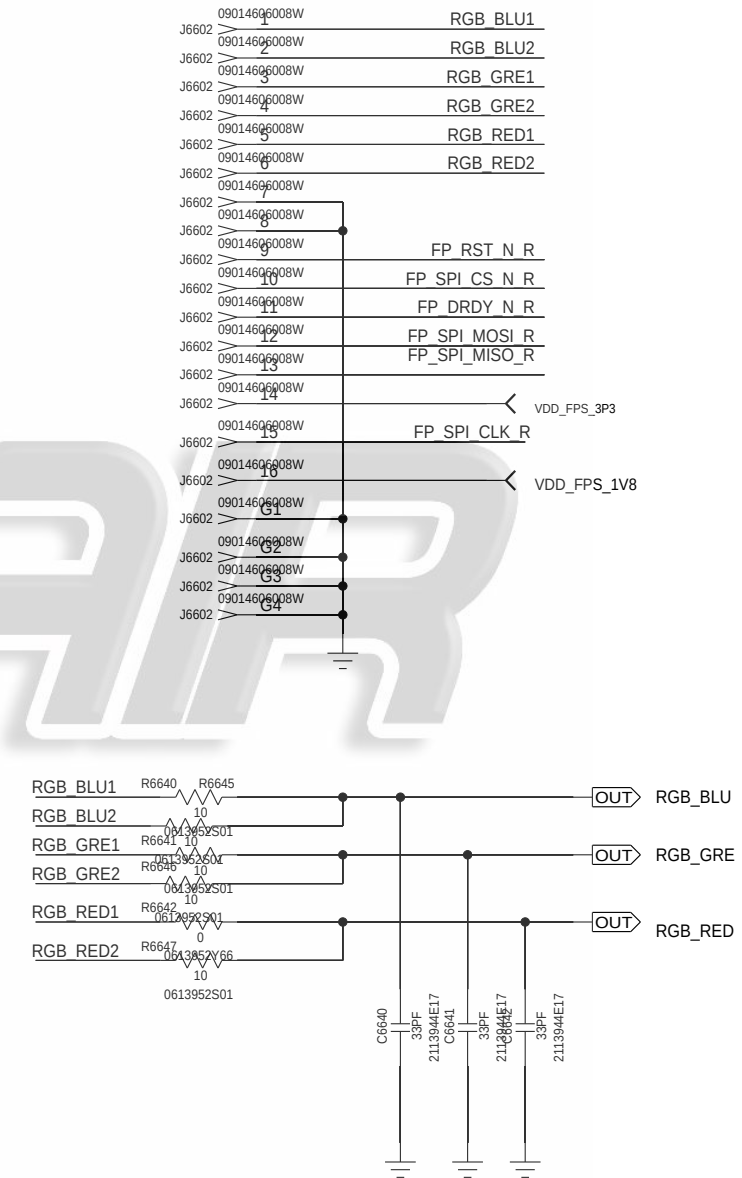
MECHANICAL KEYS

VOL_UP/DOWN/GND TPS NEED P0 BRINGUP ONLY

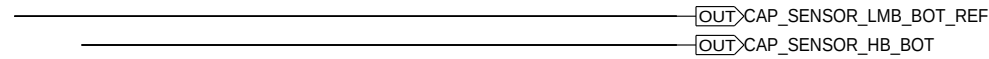
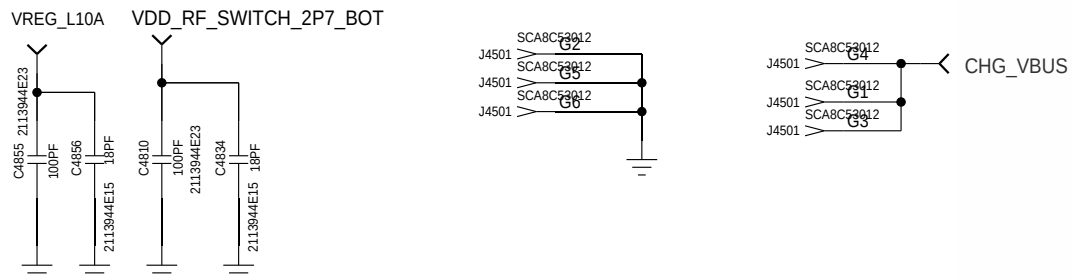
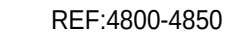


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LAST MODIFIED BY: NATE CONNELL Tue Apr 30 14:42:04 2019			
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		CAD FILE NAME: PAGE 30 OF 69	

REF: 6600 - 6699



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: NATE CONNELL	BLOCK TITLE: ui		
LAST MODIFIED BY: NATE CONNELL Tue Apr 30 14:42:04 2019			
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APPROVALS		 MOTOROLA MOBILE DEVICES		
ENGINEER: CHUNLEI YOU	BLOCK TITLE: ui			
LAST MODIFIED BY: CHUNLEI YOU Tue Apr 30 14:42:04 2019				
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Audio:MICS

REF: 5700-5789

SECONDARY MIC

REF:5750-5759

GND SHIELD, AWAY FROM NOISY SIGNAL

SEC_MIC_BIAS >>

SEC_MIC_DP <<

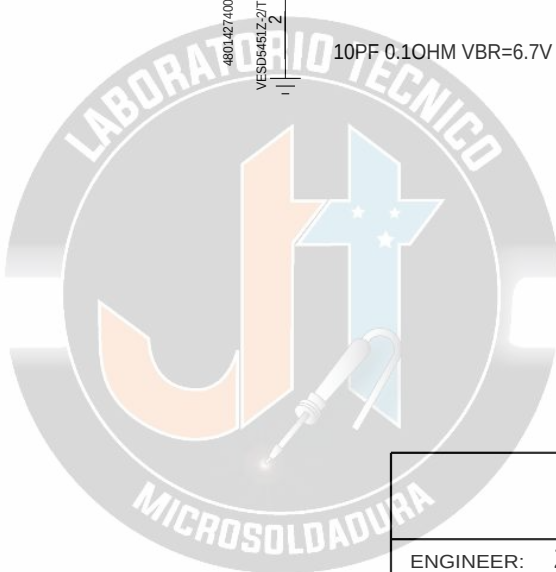
SEC_MIC_DM <<

ROUTE AS DIFF PAIR, GND SHIELD, TRY TO AWAY FROM RFPA, CHARGER, SMPS,AUDIO SMART PA,DDR

TOP MIC

E5750 LOCATED CLOSED TO MIC OUTPUT

10PF 0.1OHM VBR=6.7V MAX



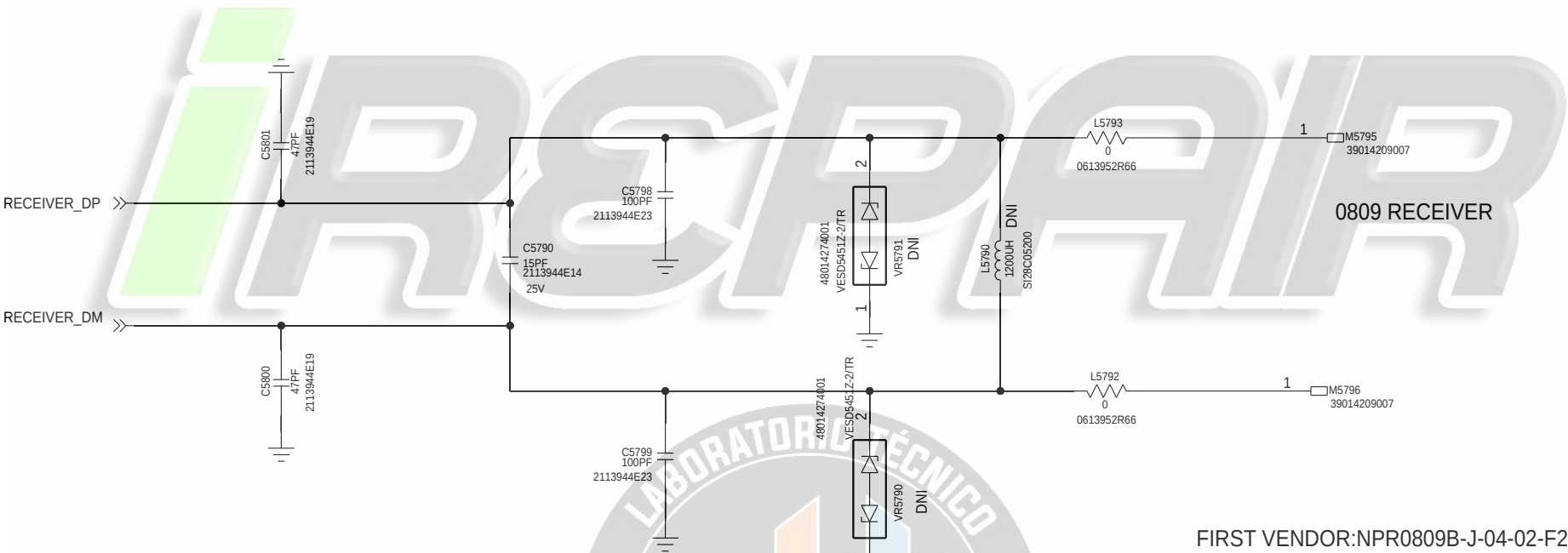
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LAST MODIFIED BY: ZHANG BEI Tue Apr 30 14:42:05 2019			
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AUDIO: Receiver

ROUTE AS DIFF PAIR
LINE WIDTH >=1MM
ADD STITCHING VIAS ALONGSIDE IF POSSIBLE

LOUDSPEARKER/EAR PIECE

REF:5790-5799



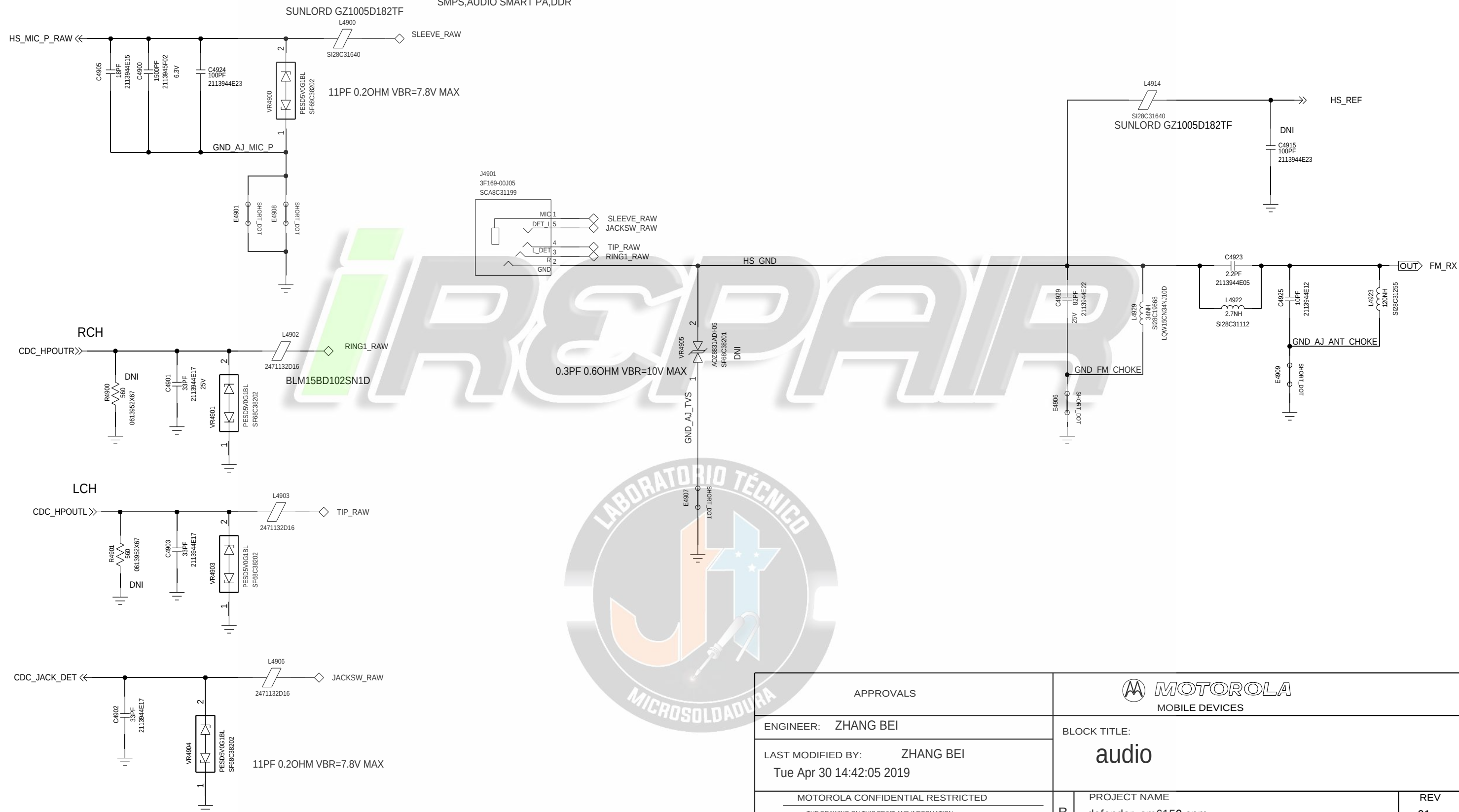
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LAST MODIFIED BY: ZHANG BEI Tue Apr 30 14:42:05 2019		audio	
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AFE: 3.5MM HEADSET

REF:4900-4949

GND SHIELD,
TRY TO AWAY FROM RFPA, CHARGER,
SMPS,AUDIO SMART PA,DDR



APPROVALS

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BLOCK TITLE:

audio

PROJECT NAME
defender_sm6150.cpm

CAD FILE NAME:

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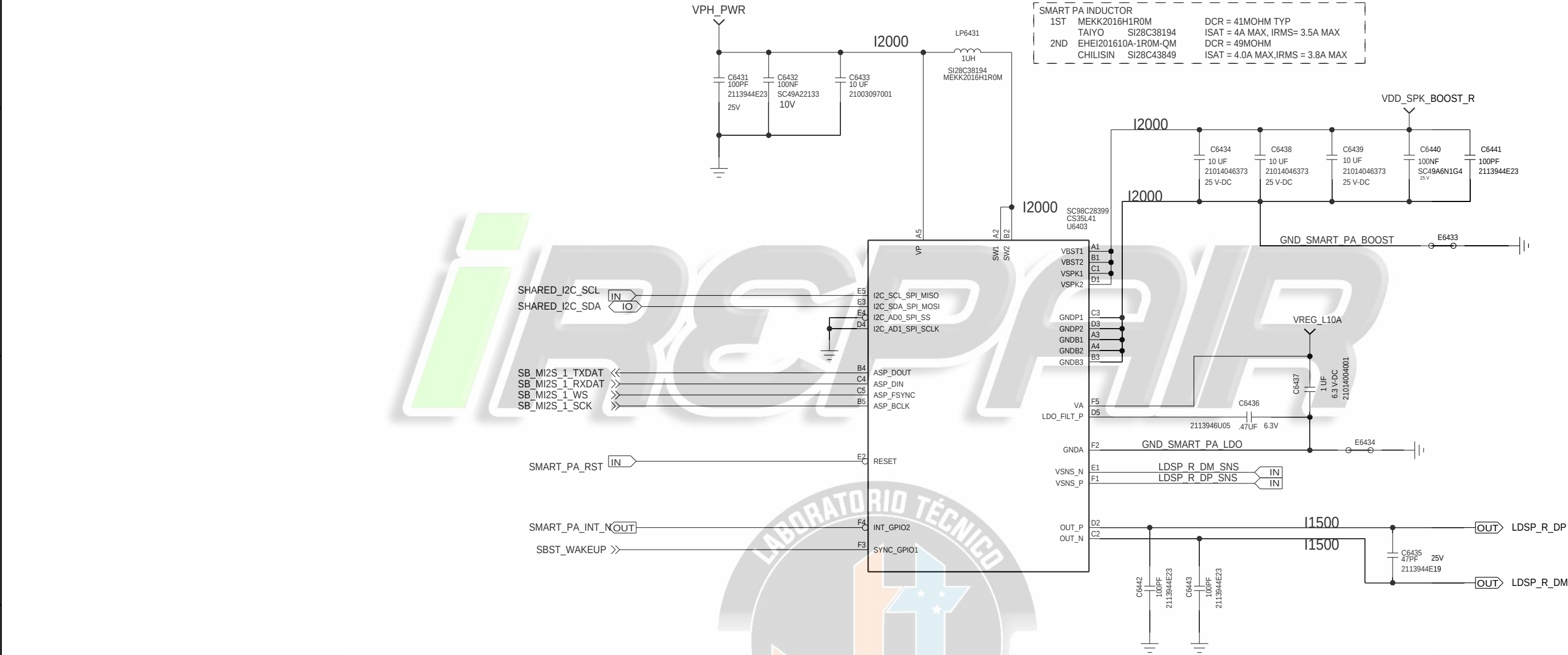
PAGE 36 OF 69

AUDIO: SMART BOOST

REF 6400-6499

*** BOOST IC I2C ADD. 0X??

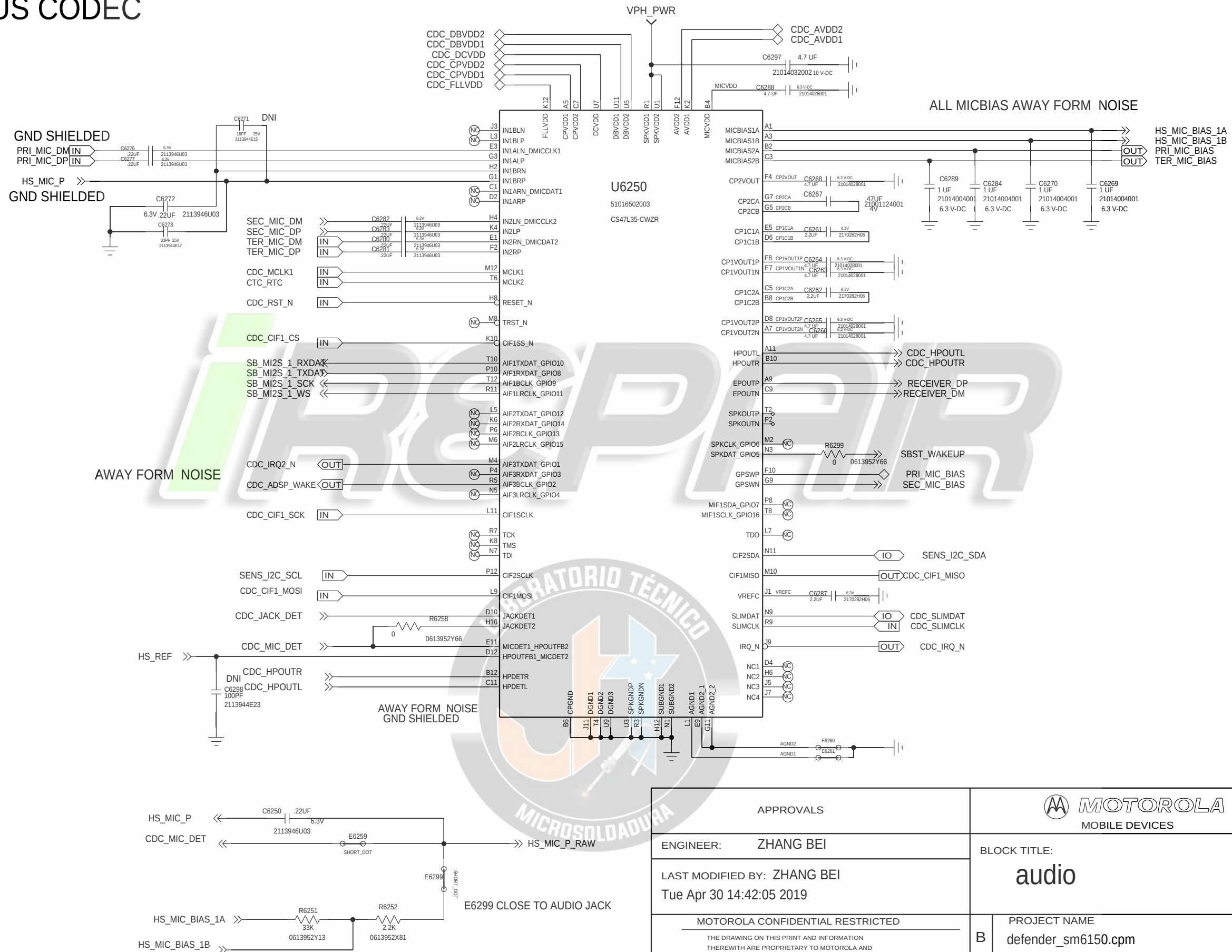
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AUDIO: CIRRUS CODEC

REF 6250-6299



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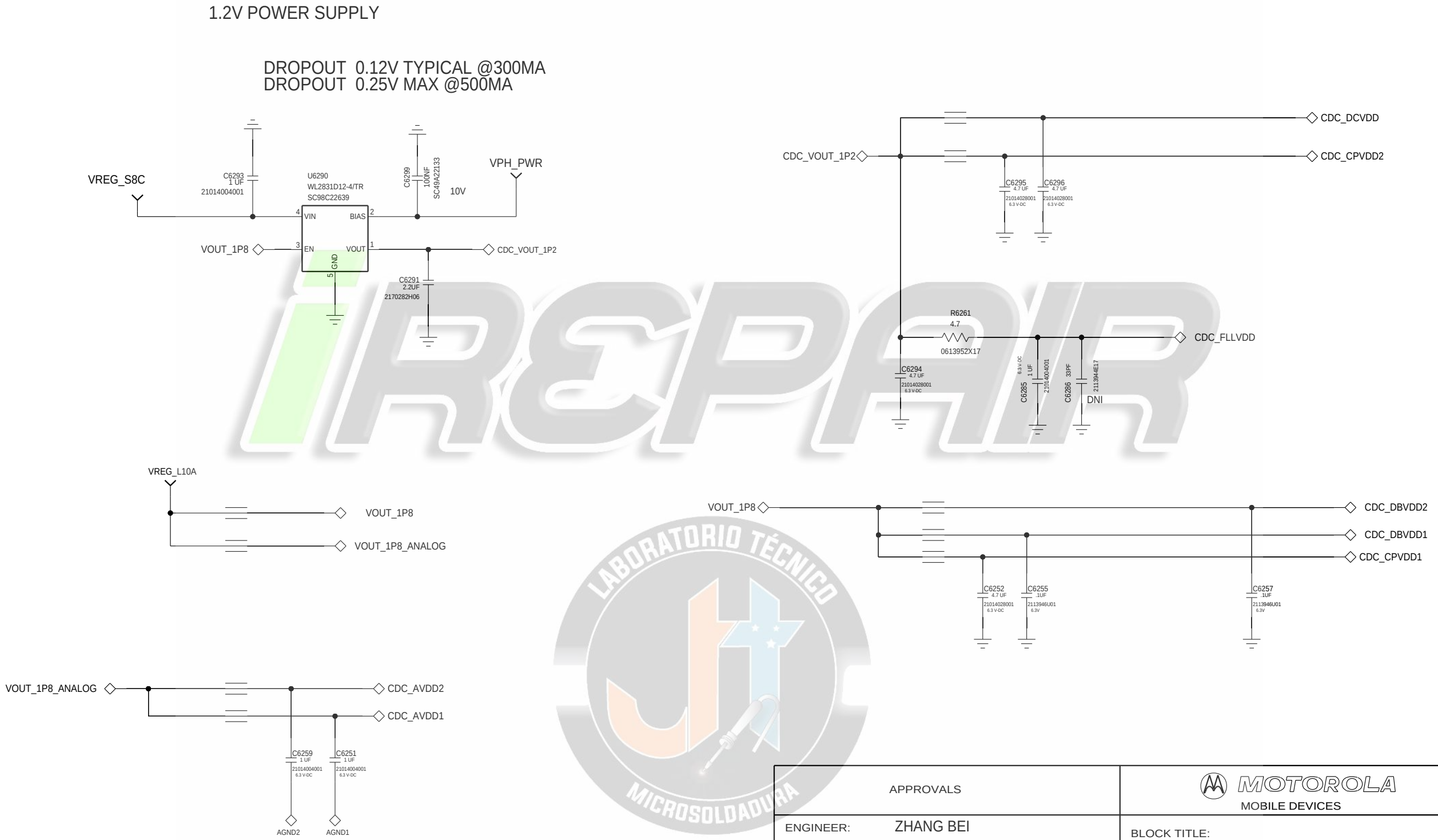
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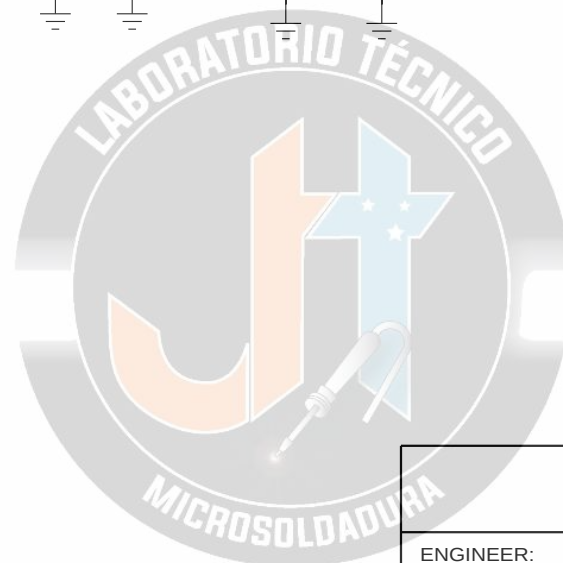
AUDIO: CODEC POWER

REF 6250-6299



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		PAGE 39 OF 69	

REF: 2300-2339

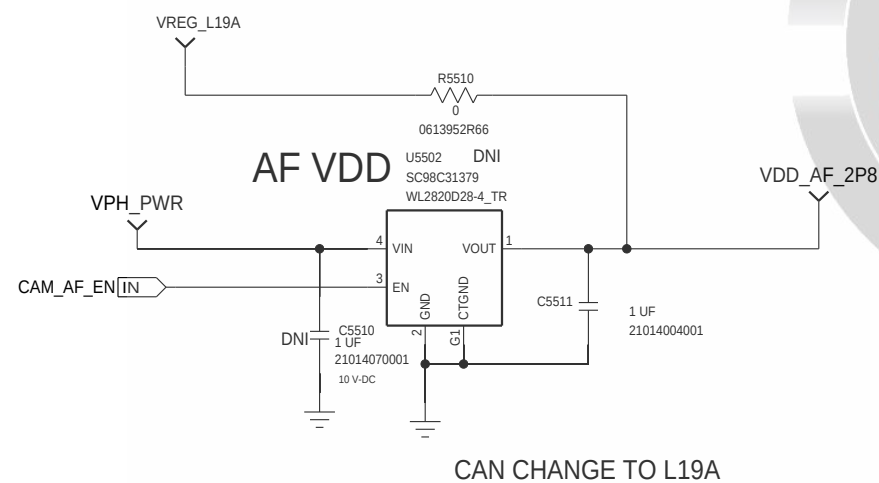
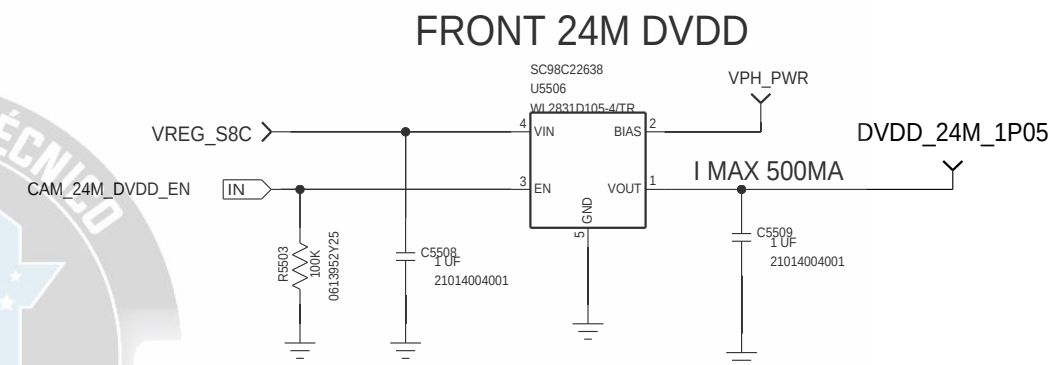
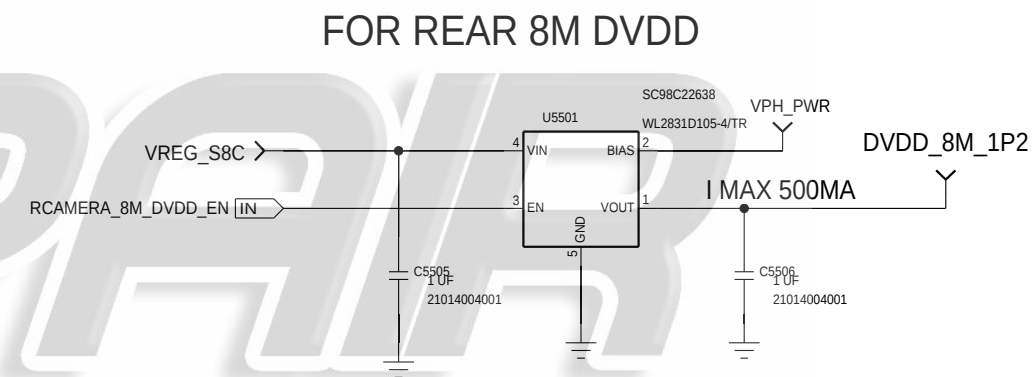
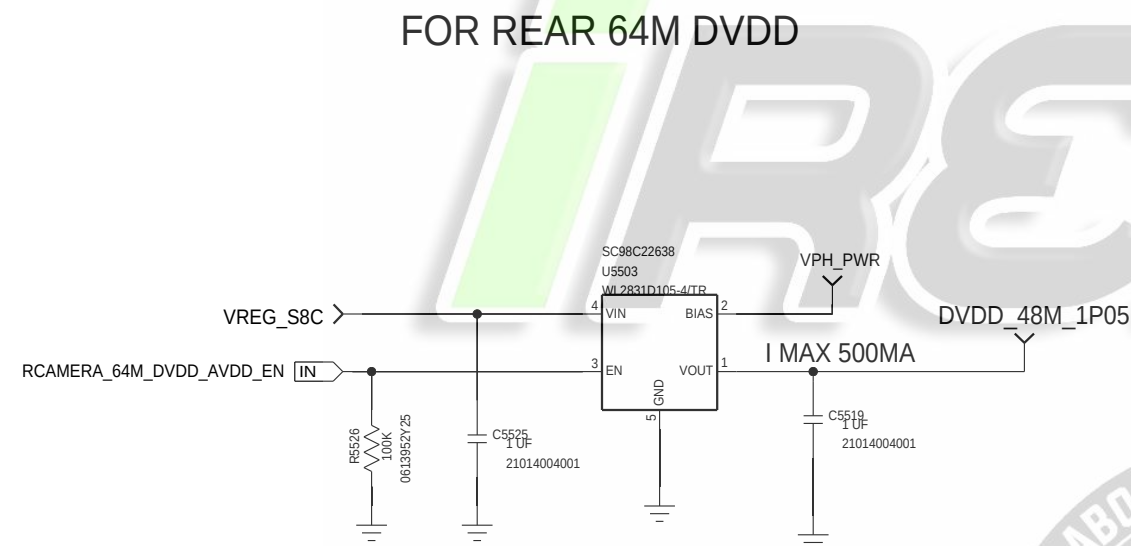
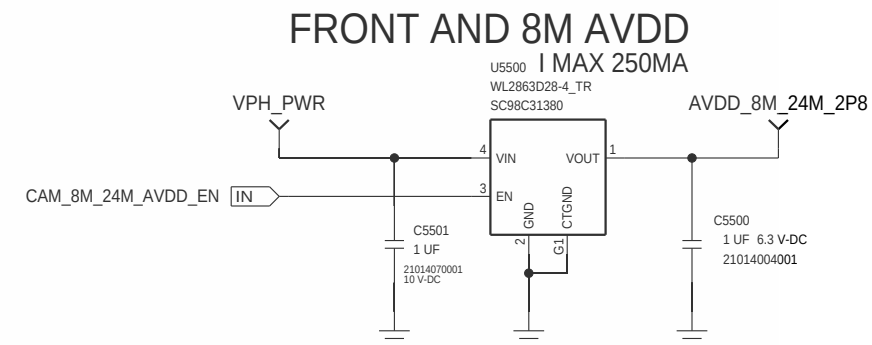
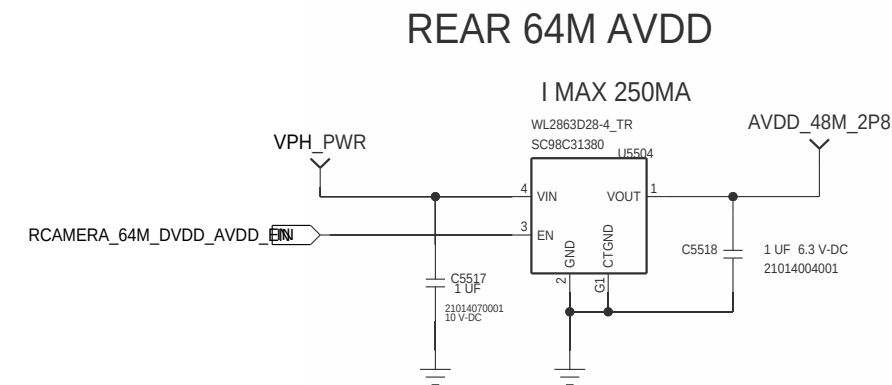
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		CAD FILE NAME:	PAGE 41 OF 69

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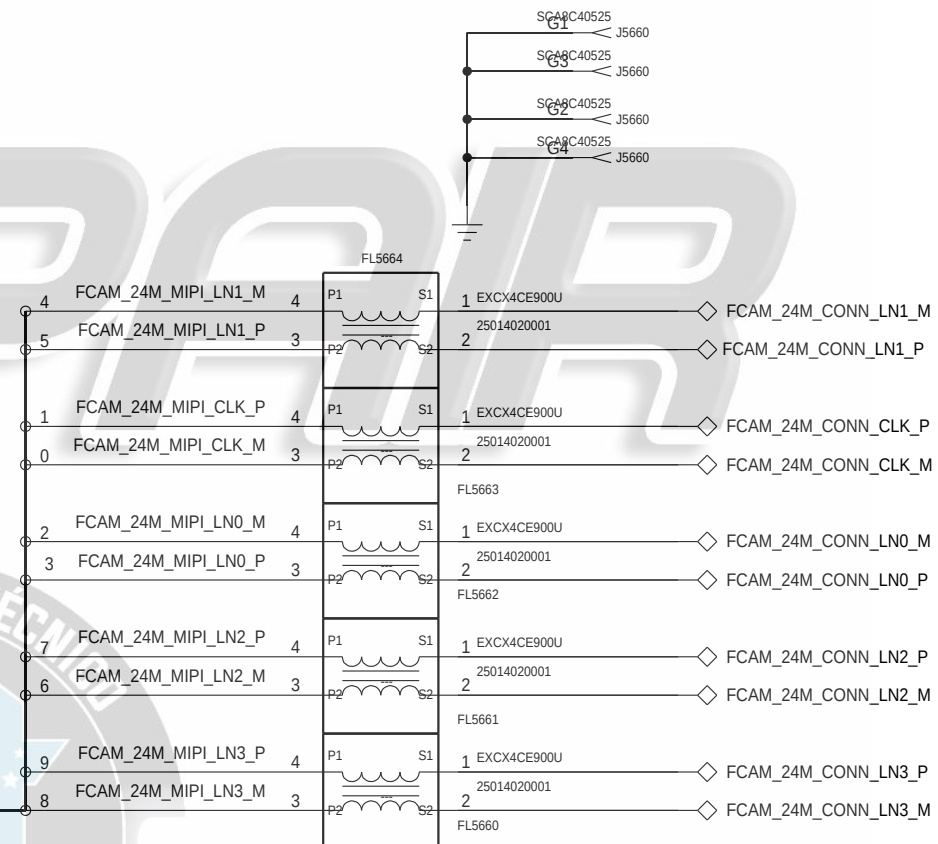
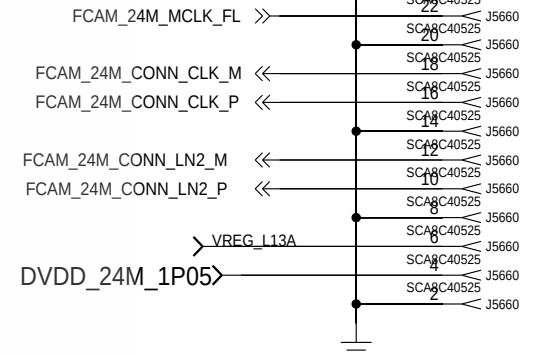
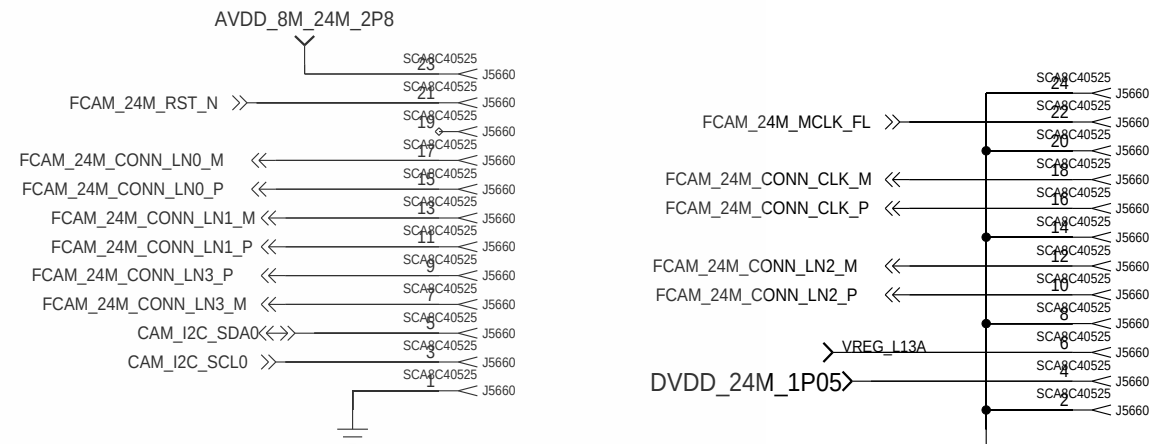
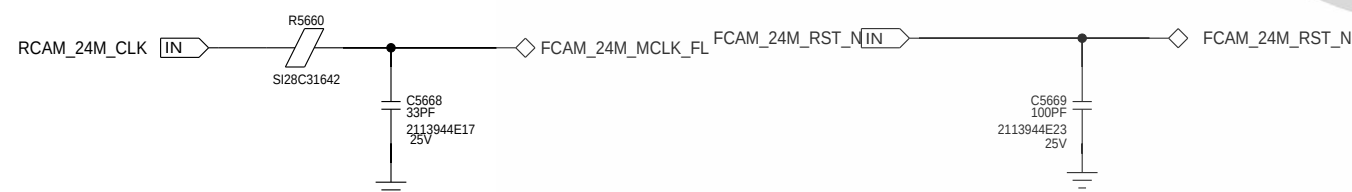
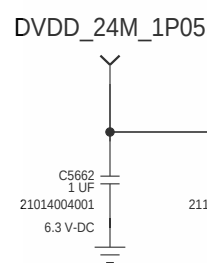
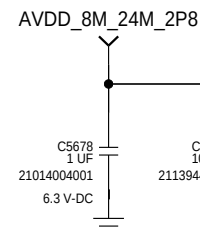
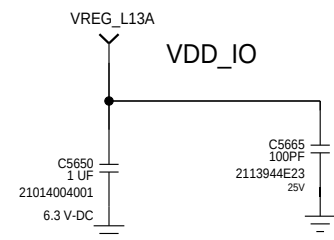
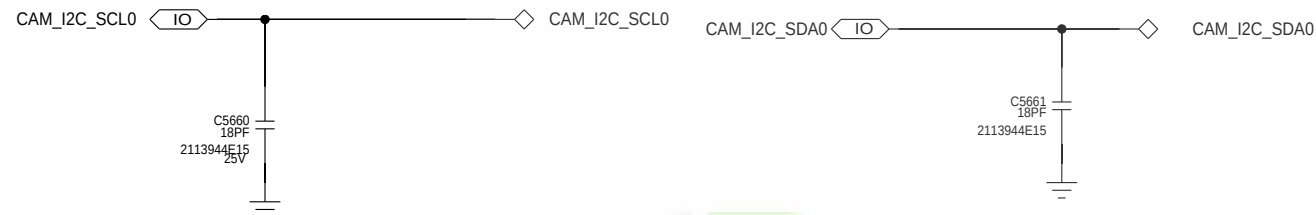


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LAST MODIFIED BY: ZHOUHU Tue Apr 30 14:42:06 2019		PROJECT NAME defender_sm6150.cpm	
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		REV 01	

CAM: FRONT 24M CONNECTOR2

REF 5660-5679

SENSOR I2C ADDR: 0X20(W), 0X21(R) FOR S5K4H7YX03-FGX9



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BLOCK TITLE:

multimedia

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PROJECT NAME
defender_sm6150.cpm

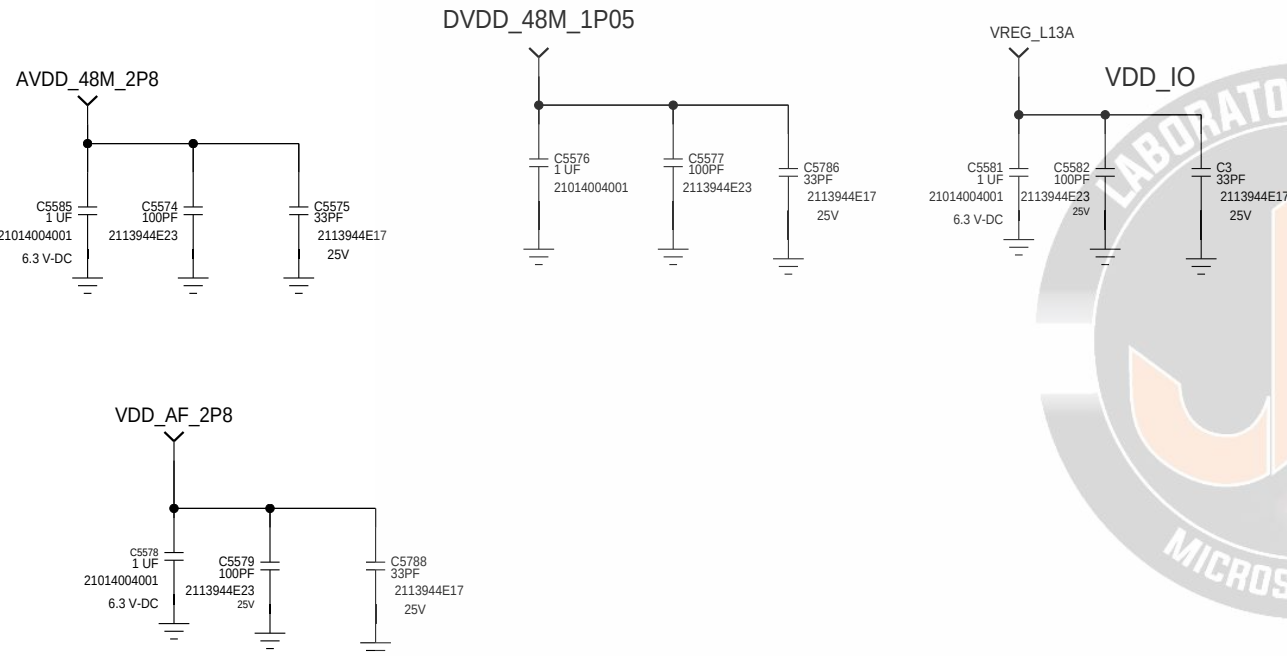
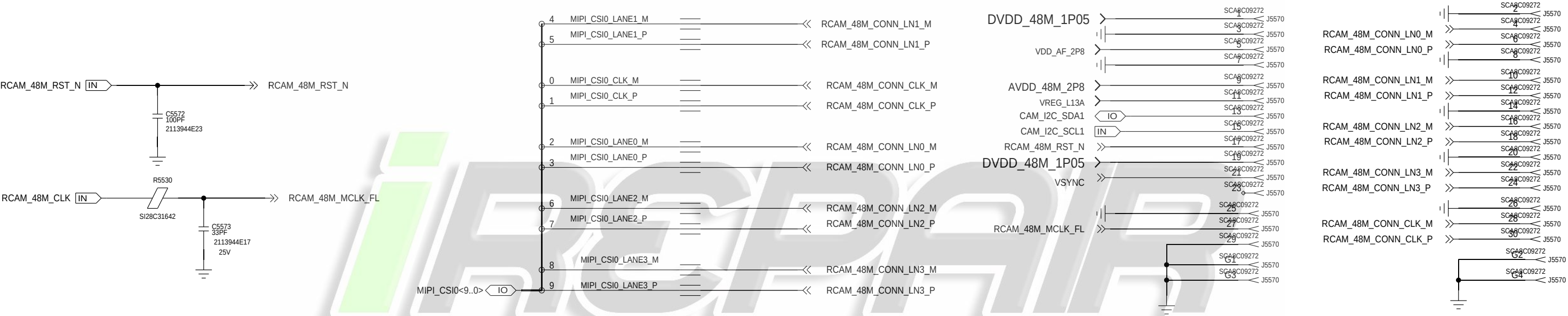
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REV
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CAM: REAR 64M MAIN CONNECTOR

REF:5570-5589



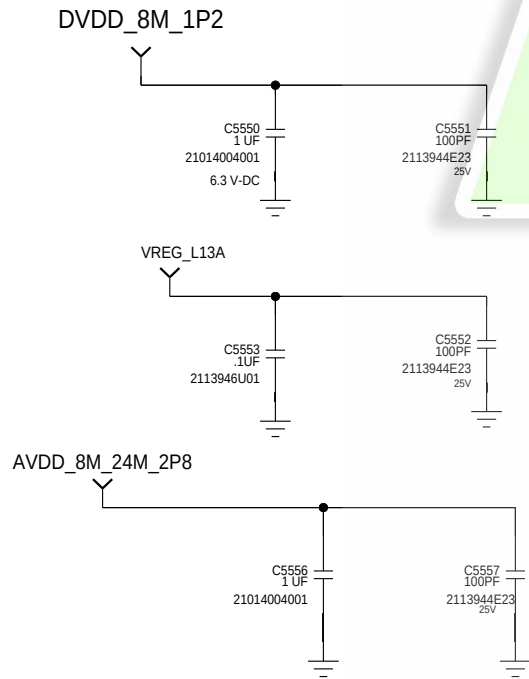
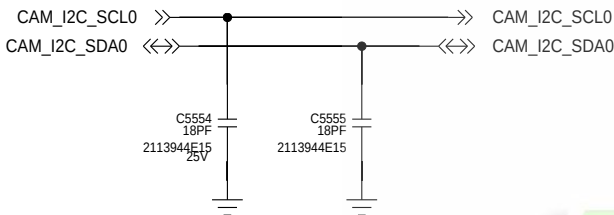
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LAST MODIFIED BY: CHUNLEI YYOU Tue Apr 30 14:42:06 2019		multimedia	
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		PAGE 44 OF 69	

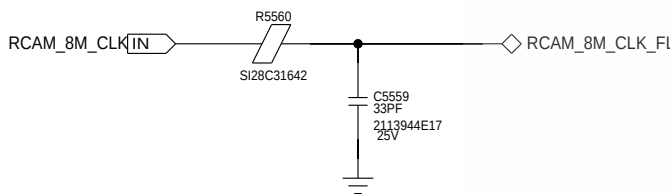
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REF 5550-5569

NEED TO CONFIRM THE PIN DIFINE

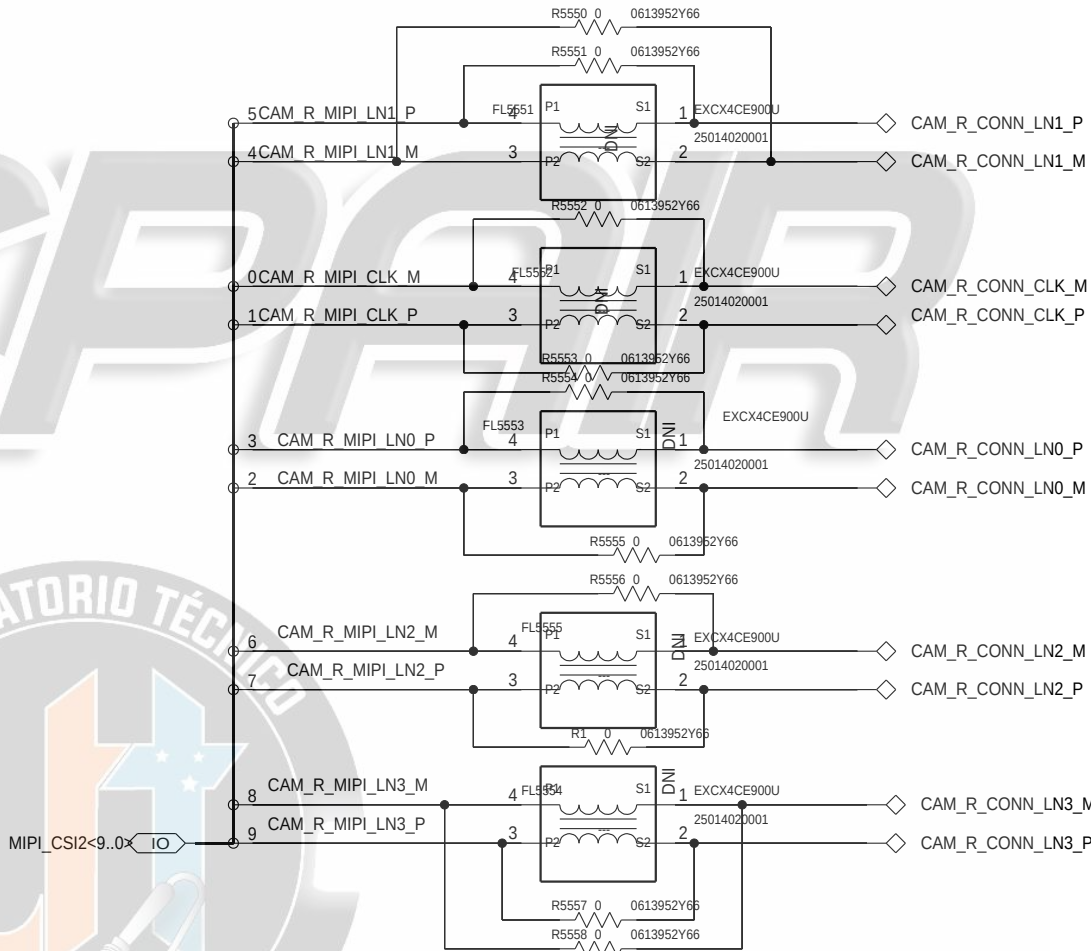
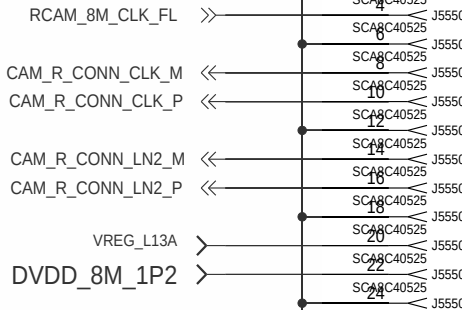
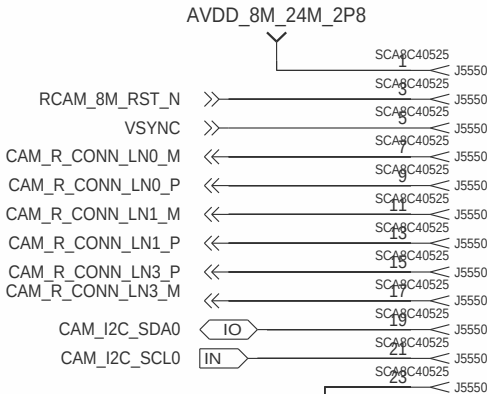
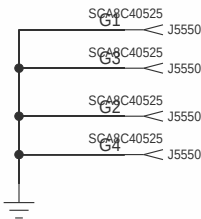
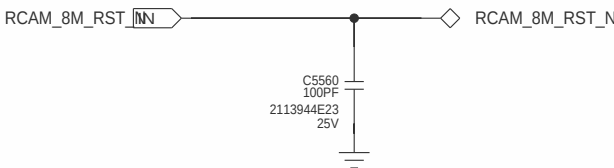
8M_I2C



EMI/ESD FILTER



XSHUTDOWN



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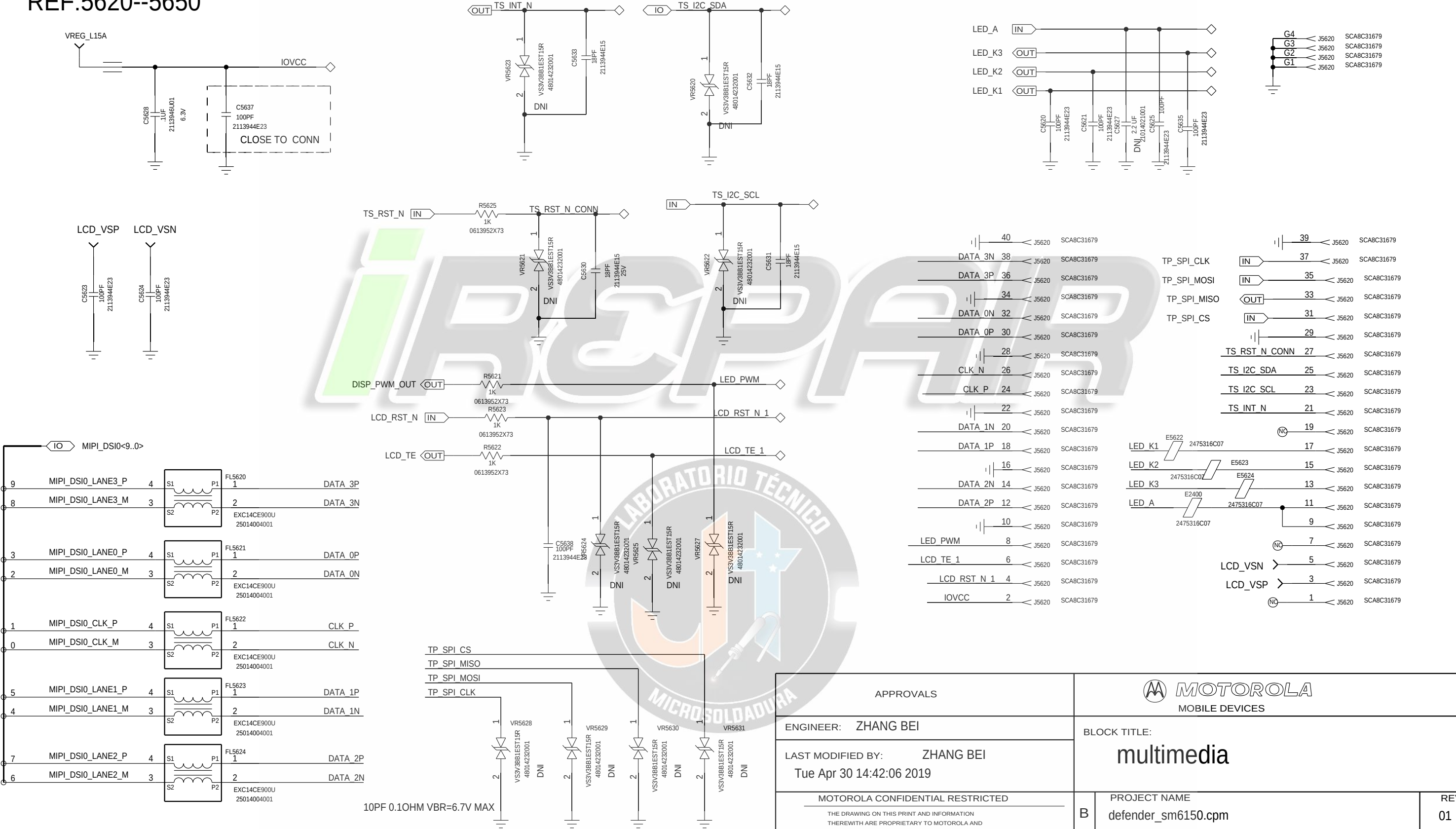
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CAD FILE NAME:

REV
01

DISP: CONNECTOR

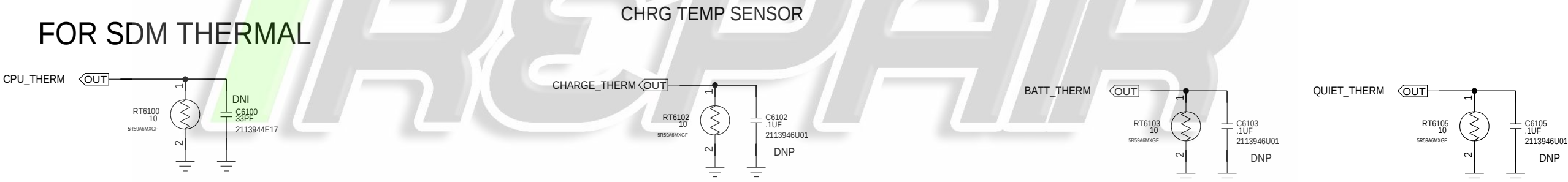
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GND SHIELDED FOR MIPI DSI SIGNALS AS A GROUP
ADD GND STITCHING VIAS FOR EACH SIGNAL GROUP (EBI,CSI,DSI) WHEN POSSIBLE TO MINIMIZE RETURN CURRENT LOOP

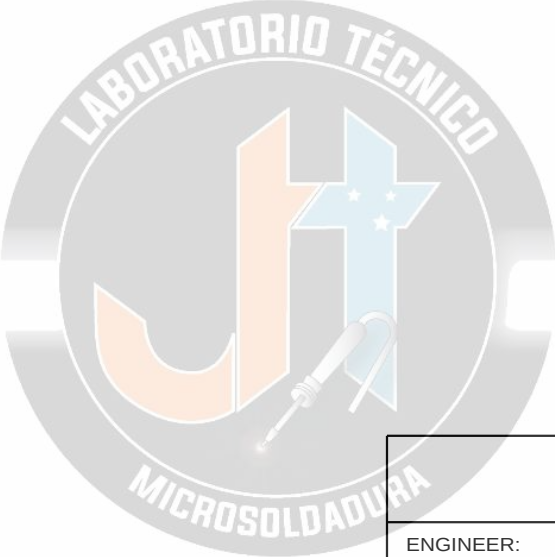
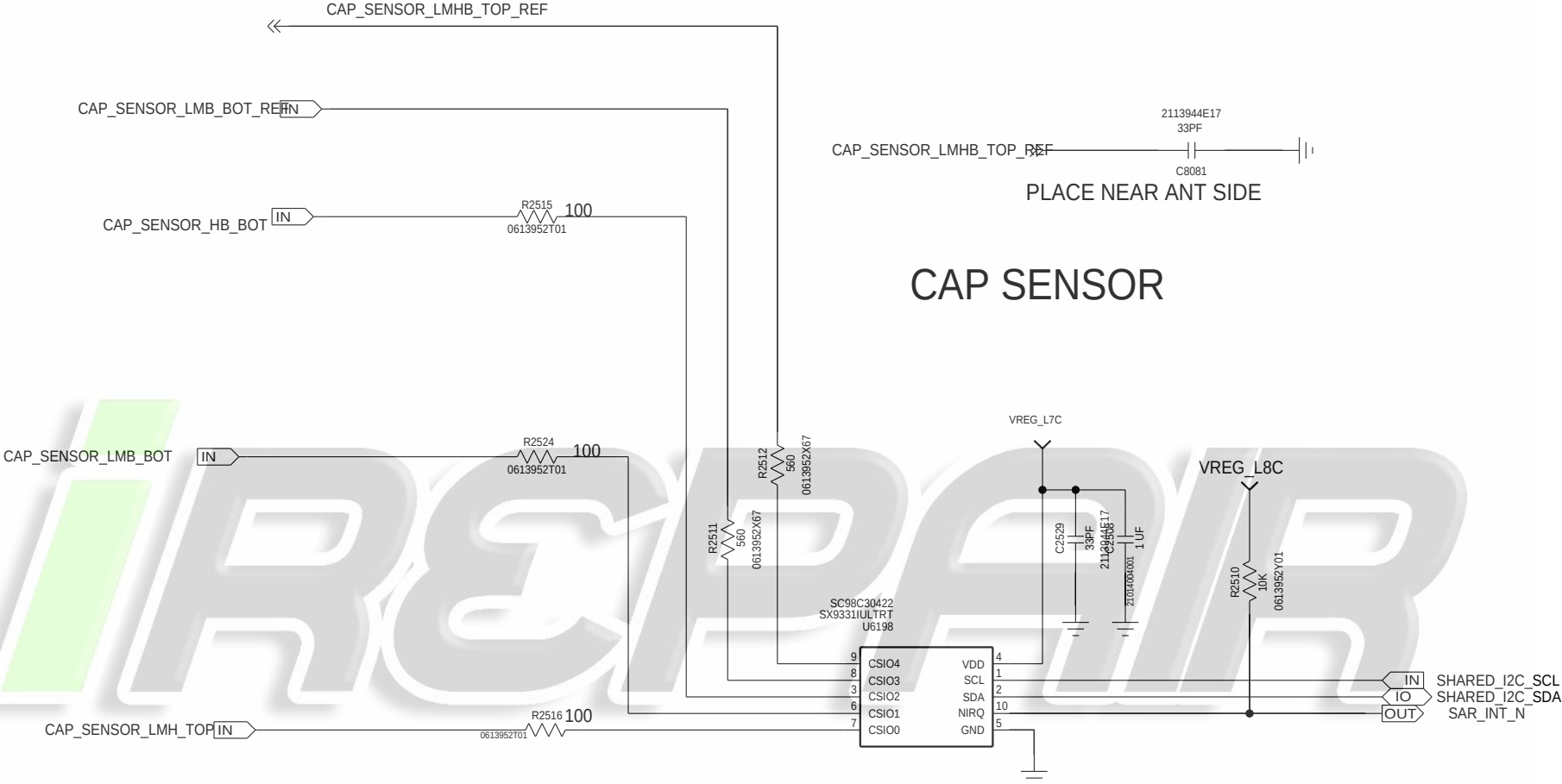
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ENGINEER: ZHANG BEI		BLOCK TITLE:	
LAST MODIFIED BY: ZHANG BEI Tue Apr 30 14:42:06 2019		multimedia	
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		CAD FILE NAME:	PAGE 46 OF 69

Sensor: NTC
REF:6100--6110
RESERVE: 6111-6149



JOHN GORSICA/NATE CONNELL APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: ZHANG BEI		BLOCK TITLE: sensor	
LAST MODIFIED BY: ZHANG BEI Tue Apr 30 14:42:07 2019			
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		REV 01 PAGE 47 OF 69	

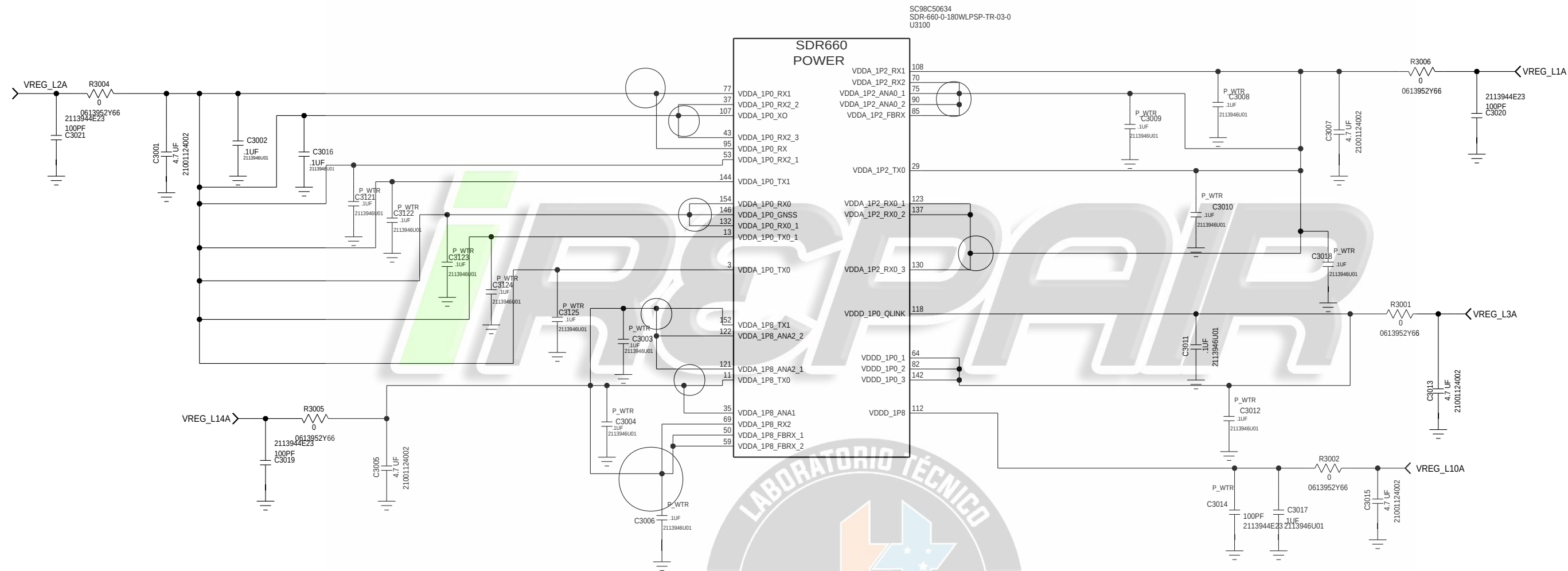
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REF 2510-2530



APPROVALS		 MOTOROLA MOBILE DEVICES	
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			REV 01 PAGE 49 OF 69

SDR660: POWER

REF 3000-3099



APPROVALS

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MOBILE DEVICES

BLOCK TITLE:

blk_sdr660

PROJECT NAME

blk_sdr660

CAD FILE NAME:

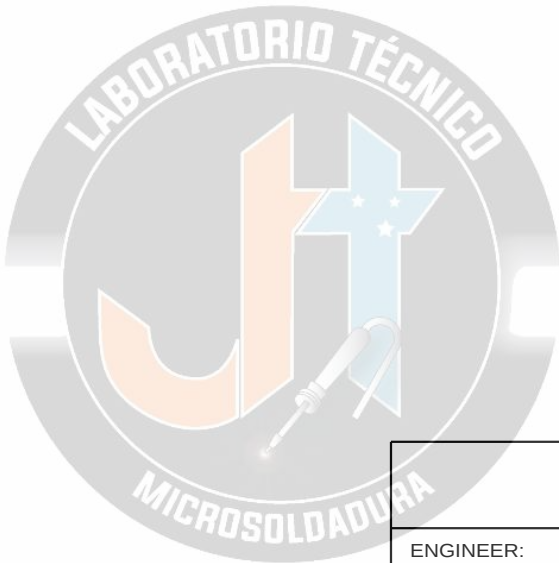
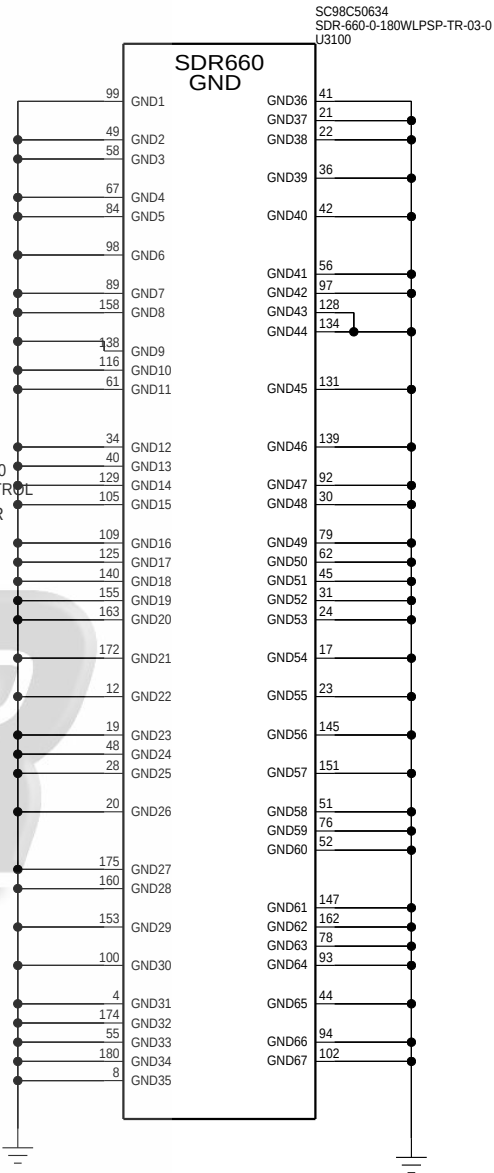
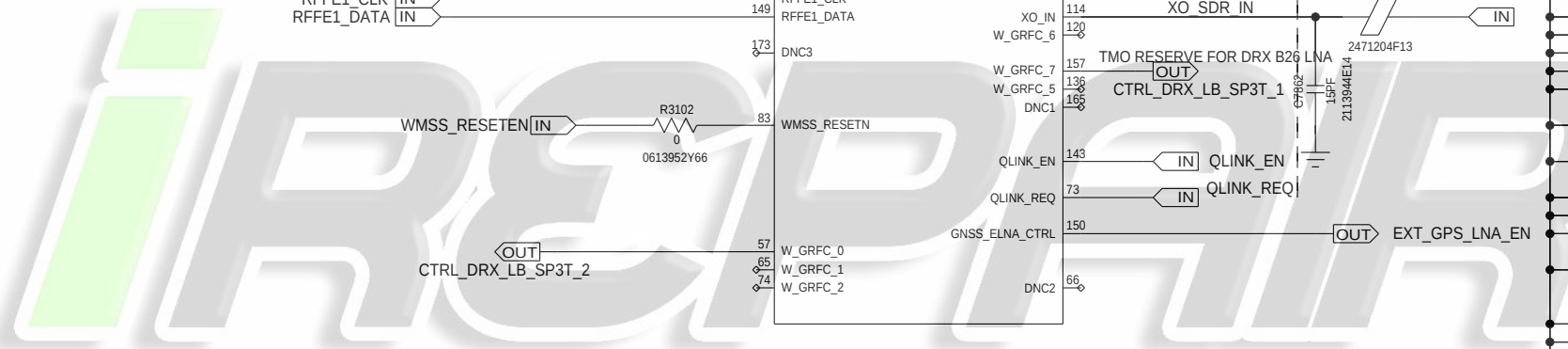
REV

01

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SDR660: TX/PWR/GROUND

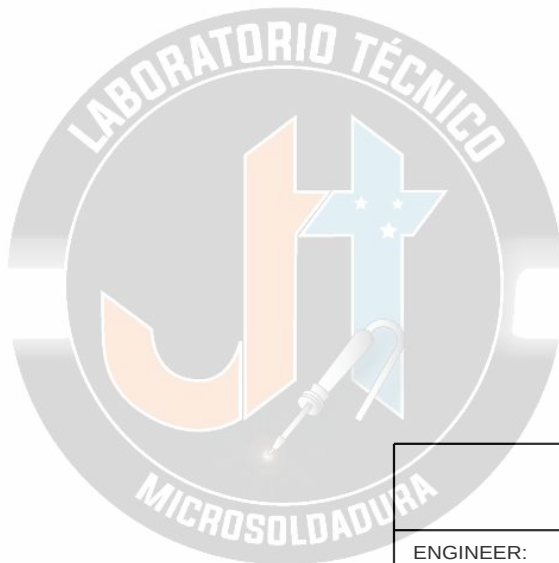
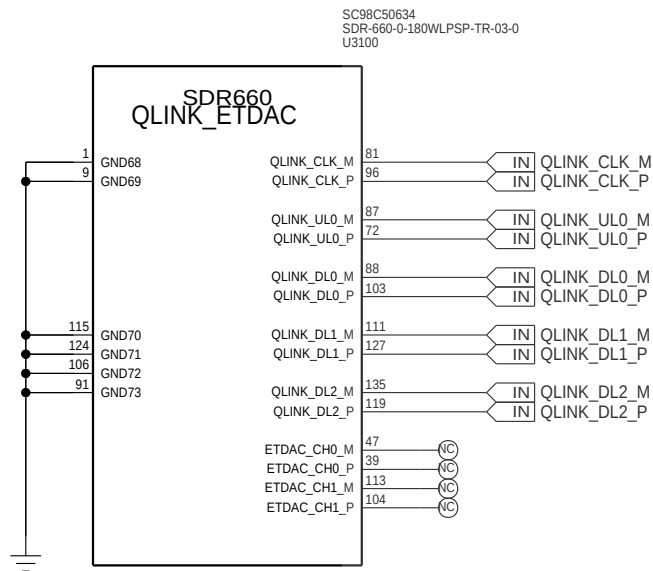
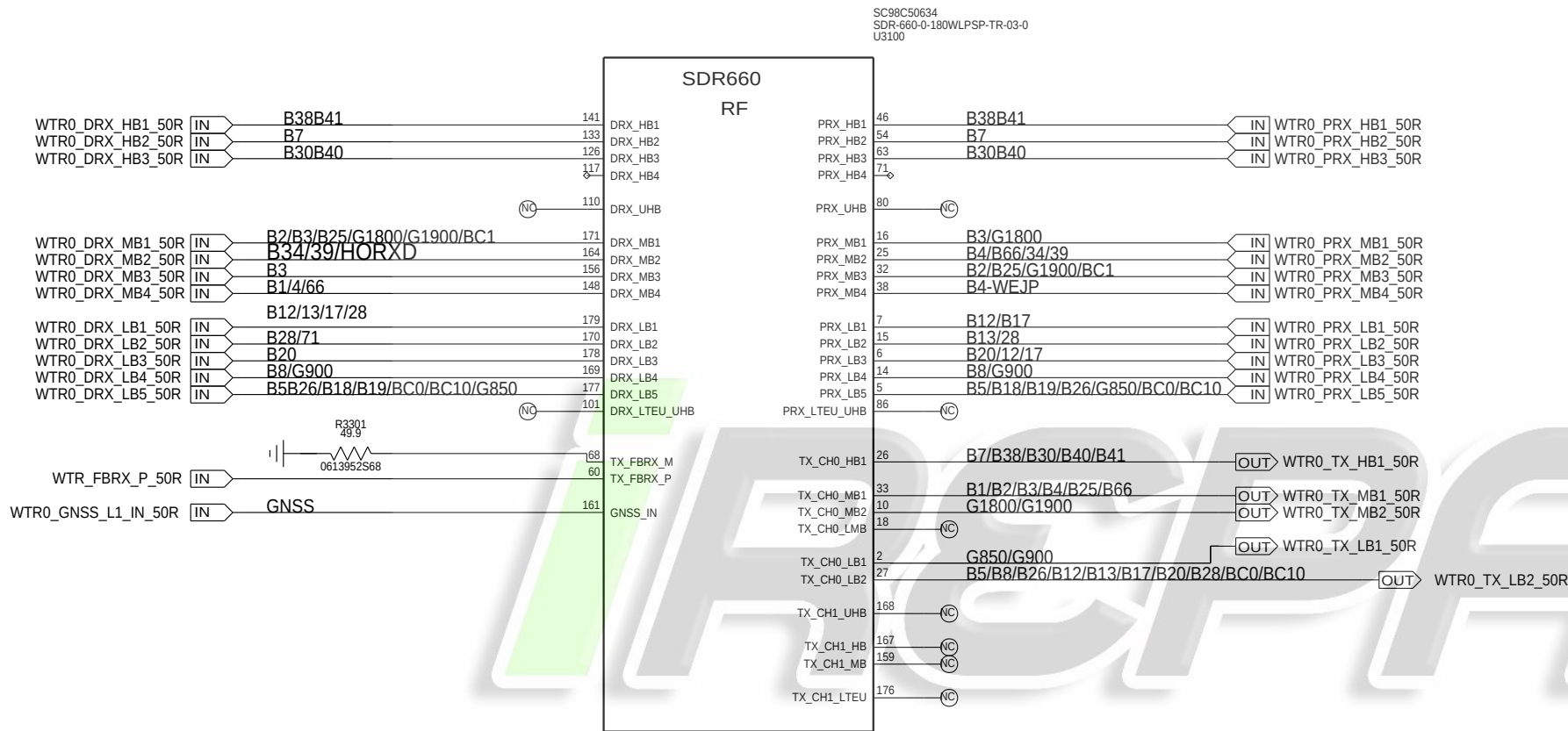
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SDR660: PRX/DRX/GPS

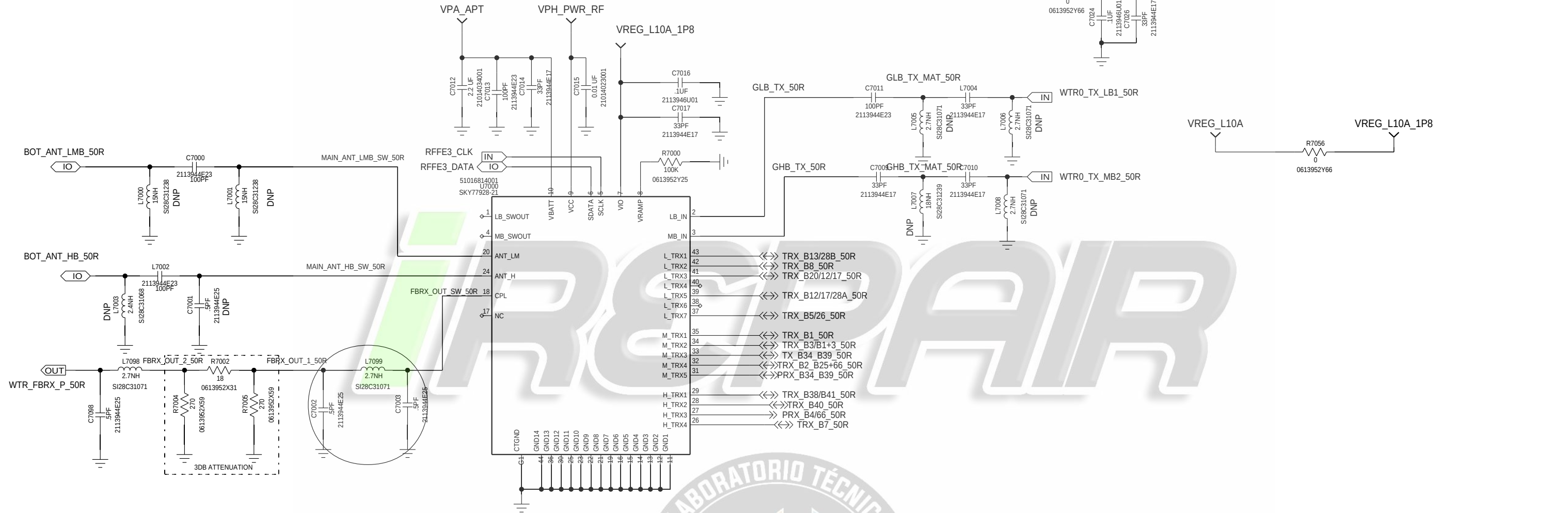
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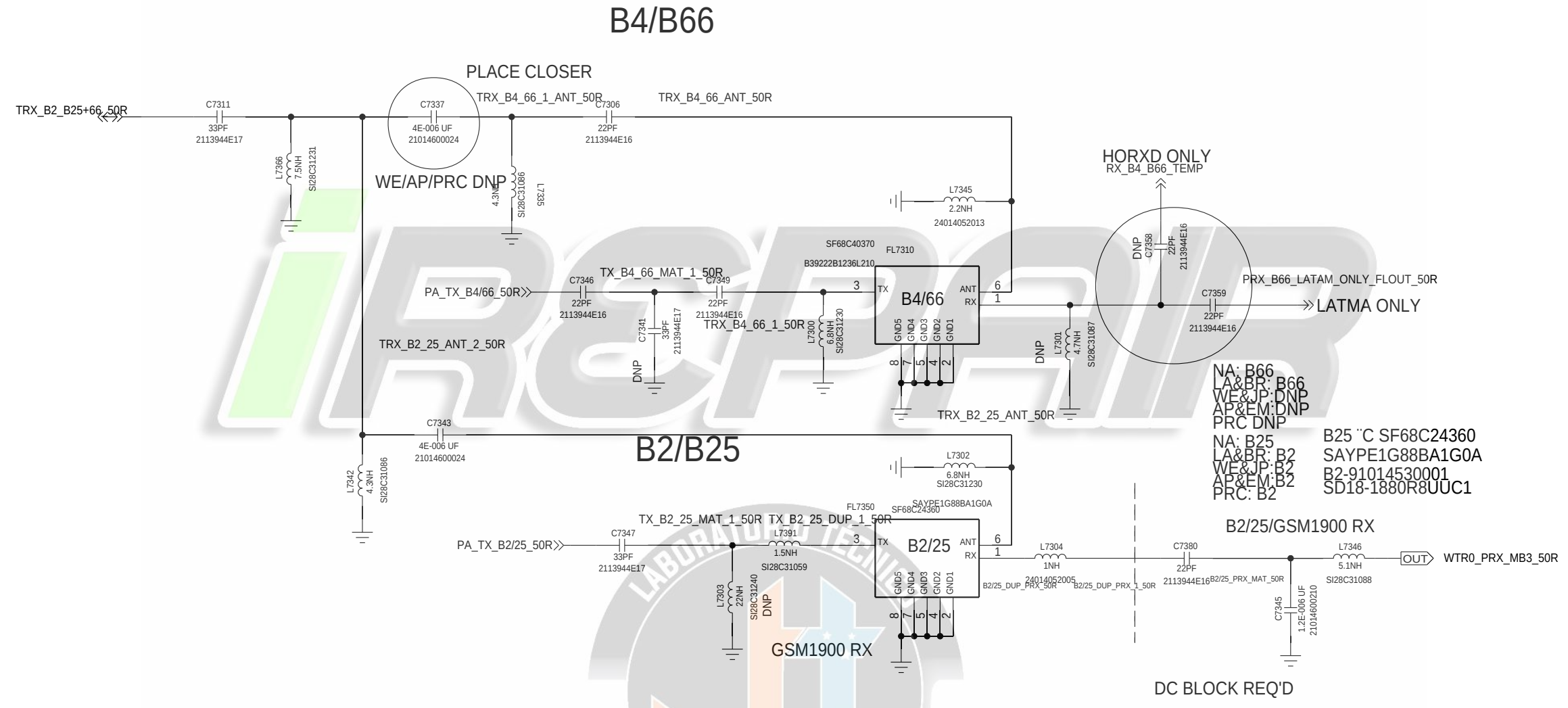
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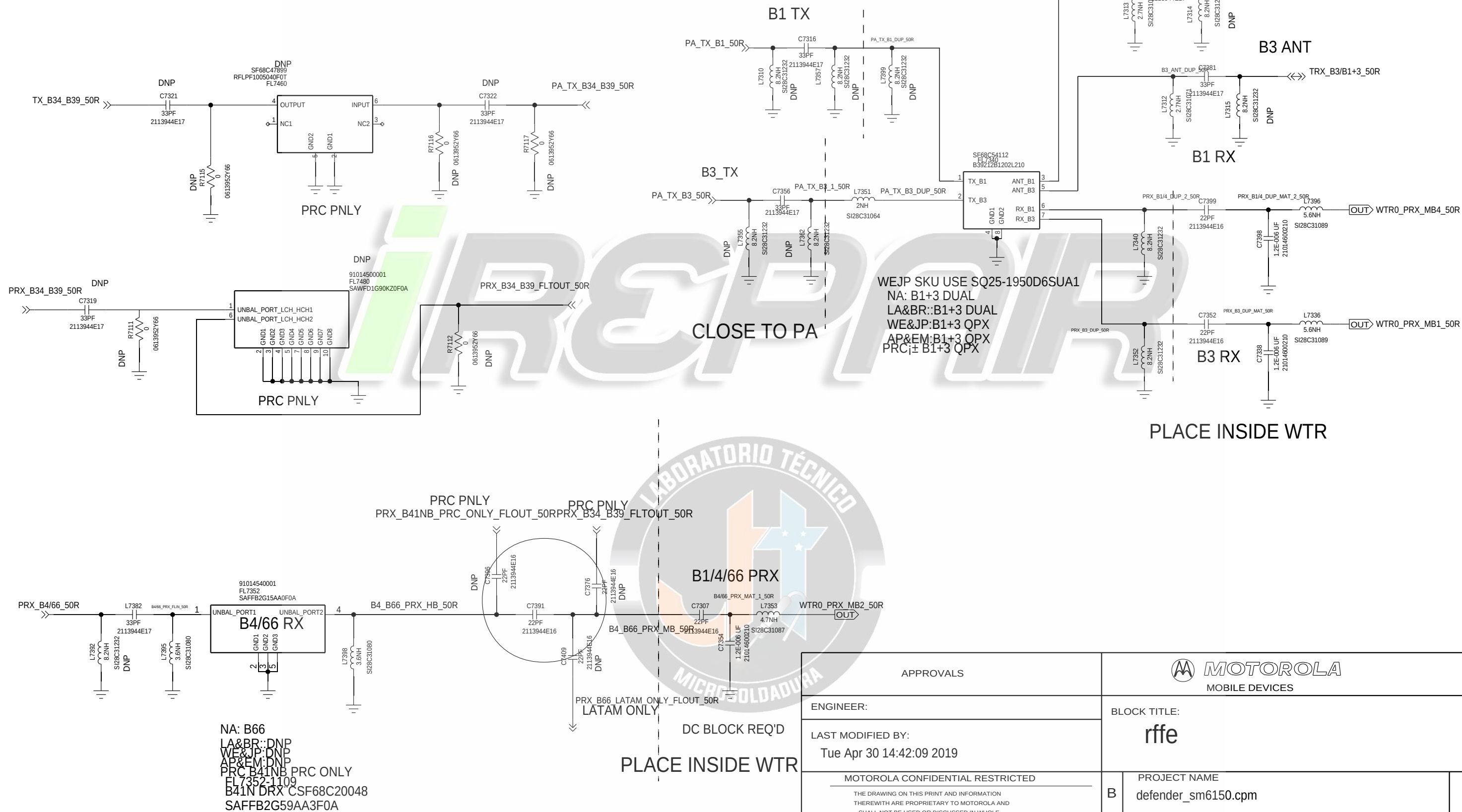
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		PAGE 56 OF 69	

RFFE: MB_TRX_B1_3_G1800,1_4_66_SINGLE_SAW

REF 7350-7399



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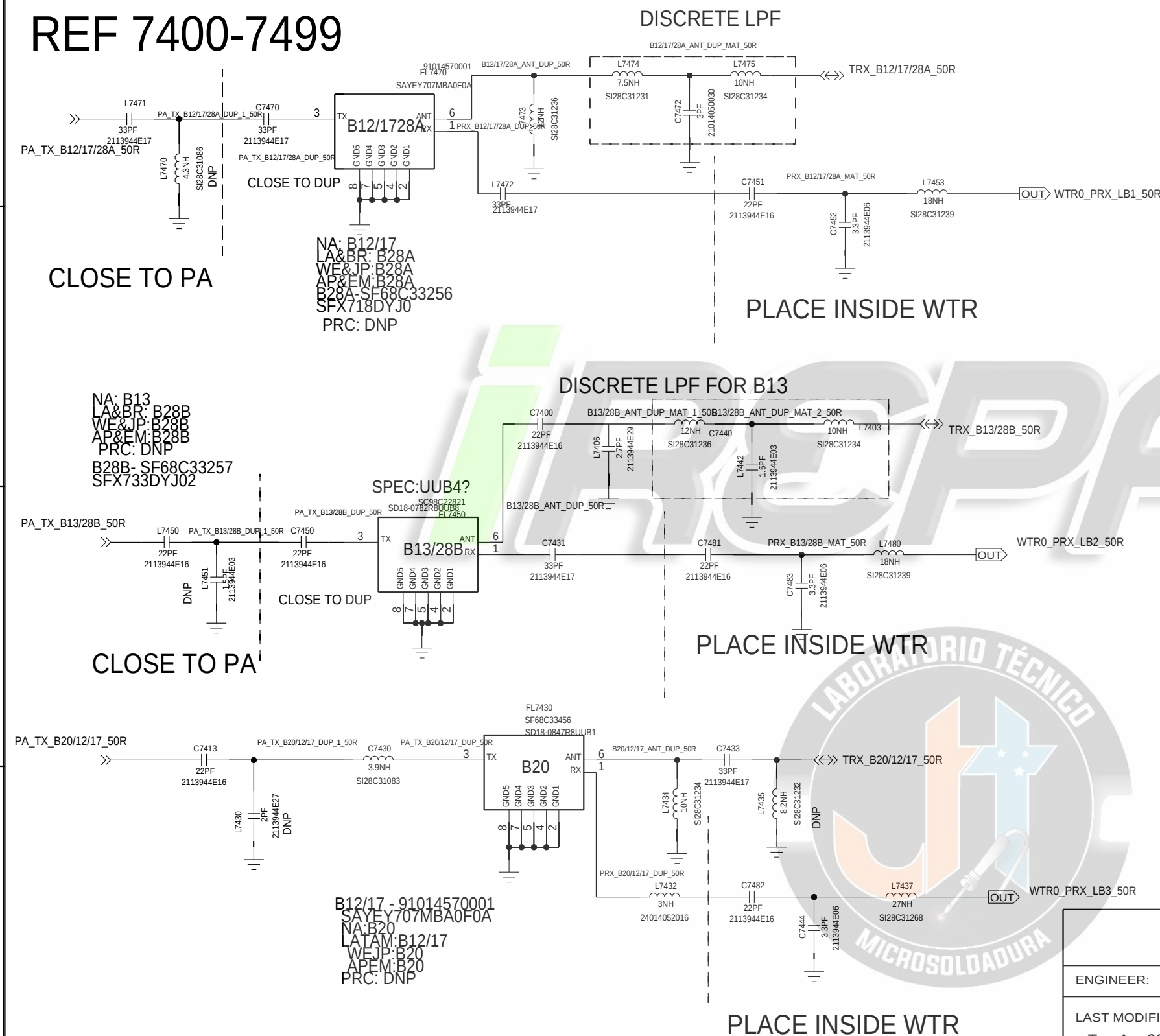
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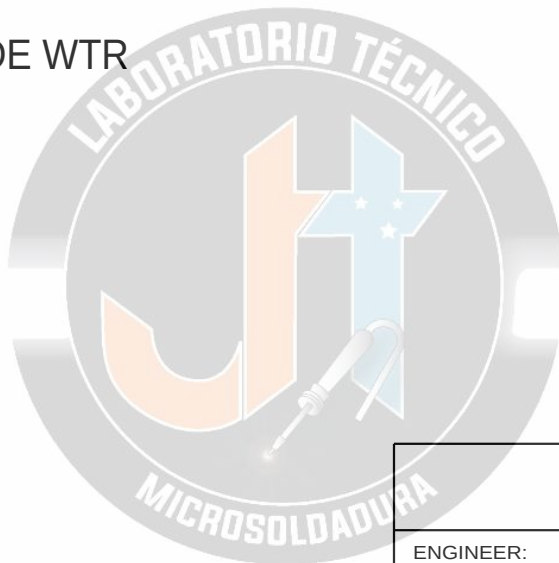
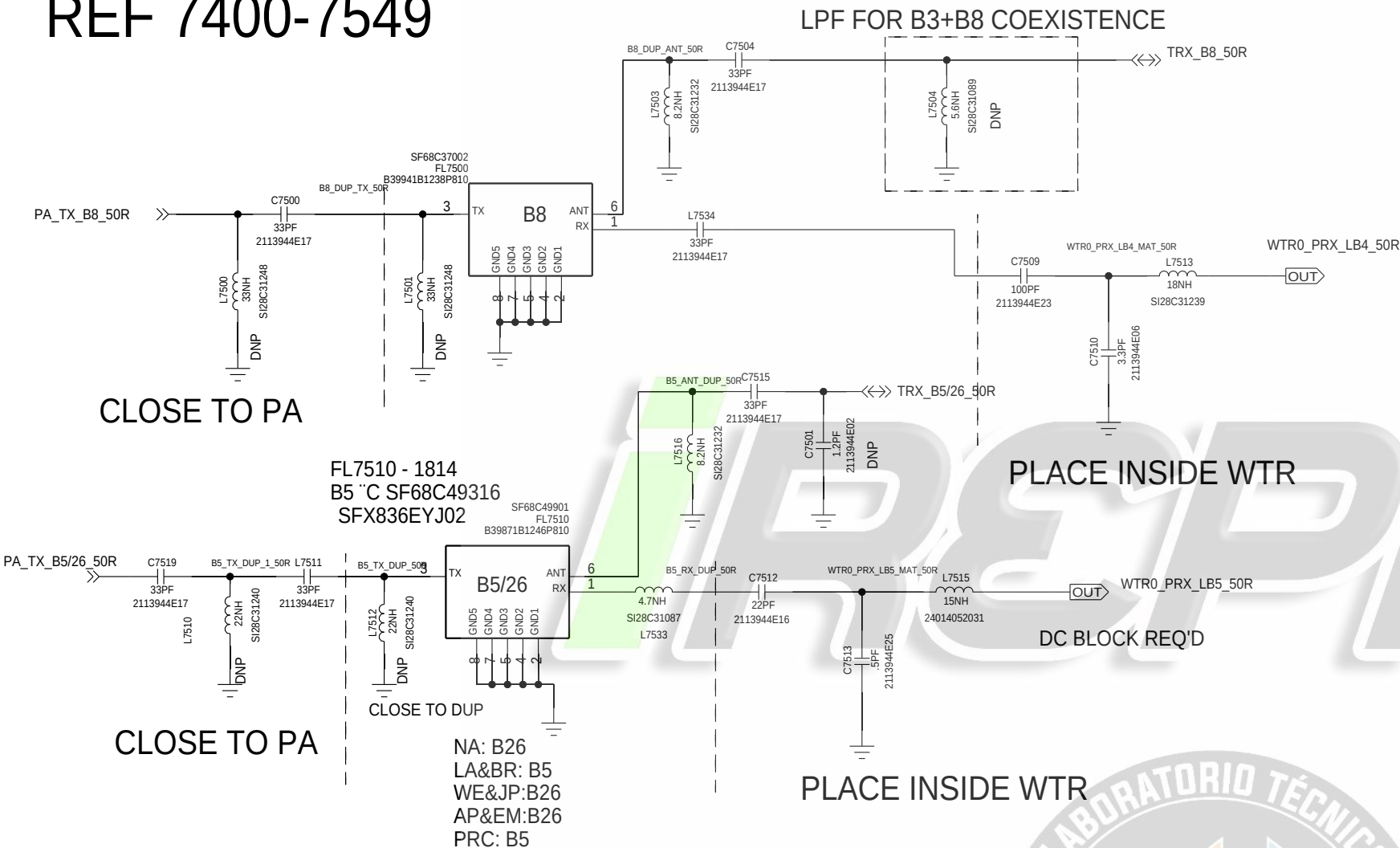
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RFFE: LB_TRX_B12,17,13,20,28AB
REF 7400-7499



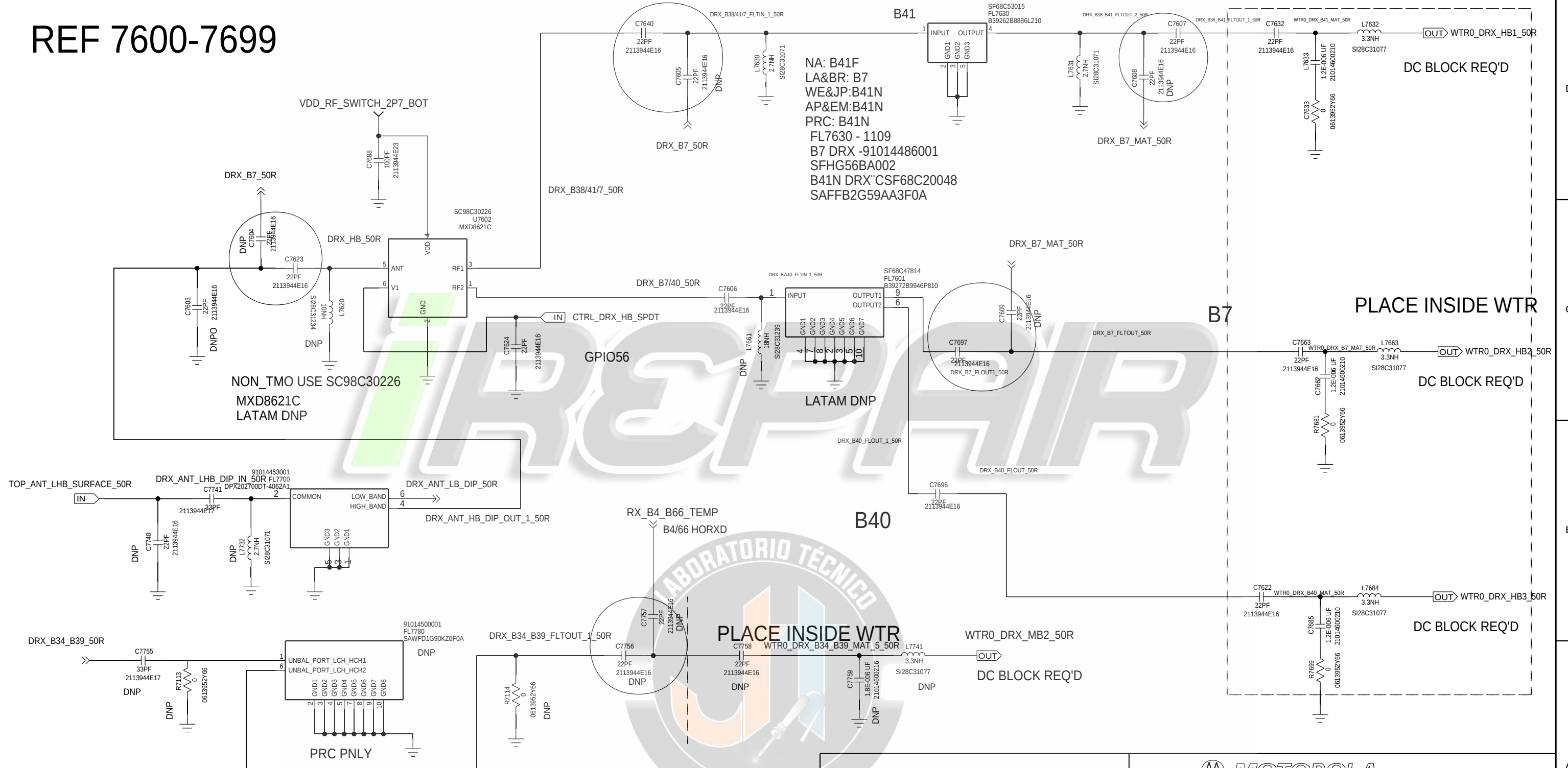
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RFFE: LB TRX B5,8,26,G850/900,71
REF 7400-7549



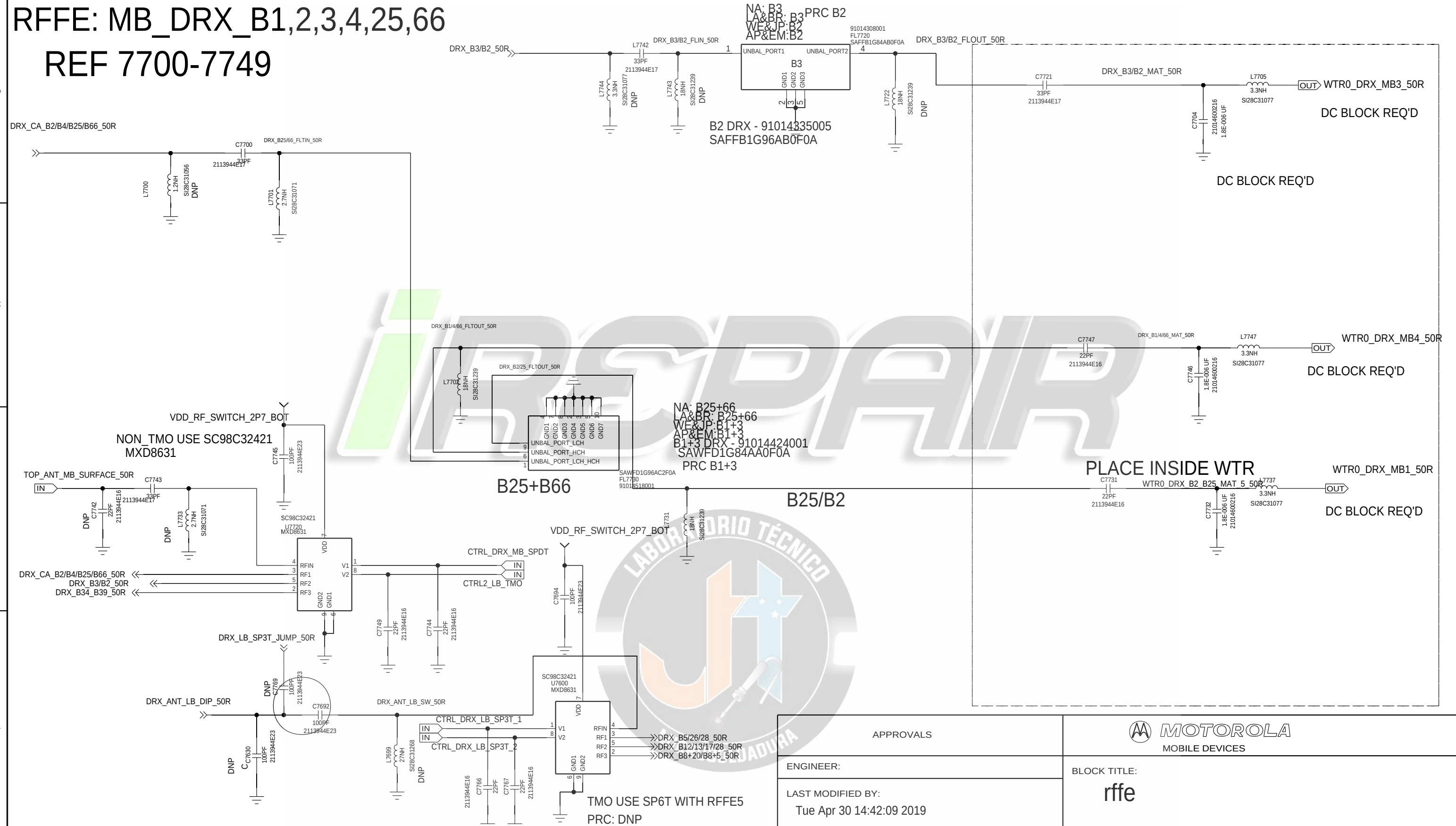
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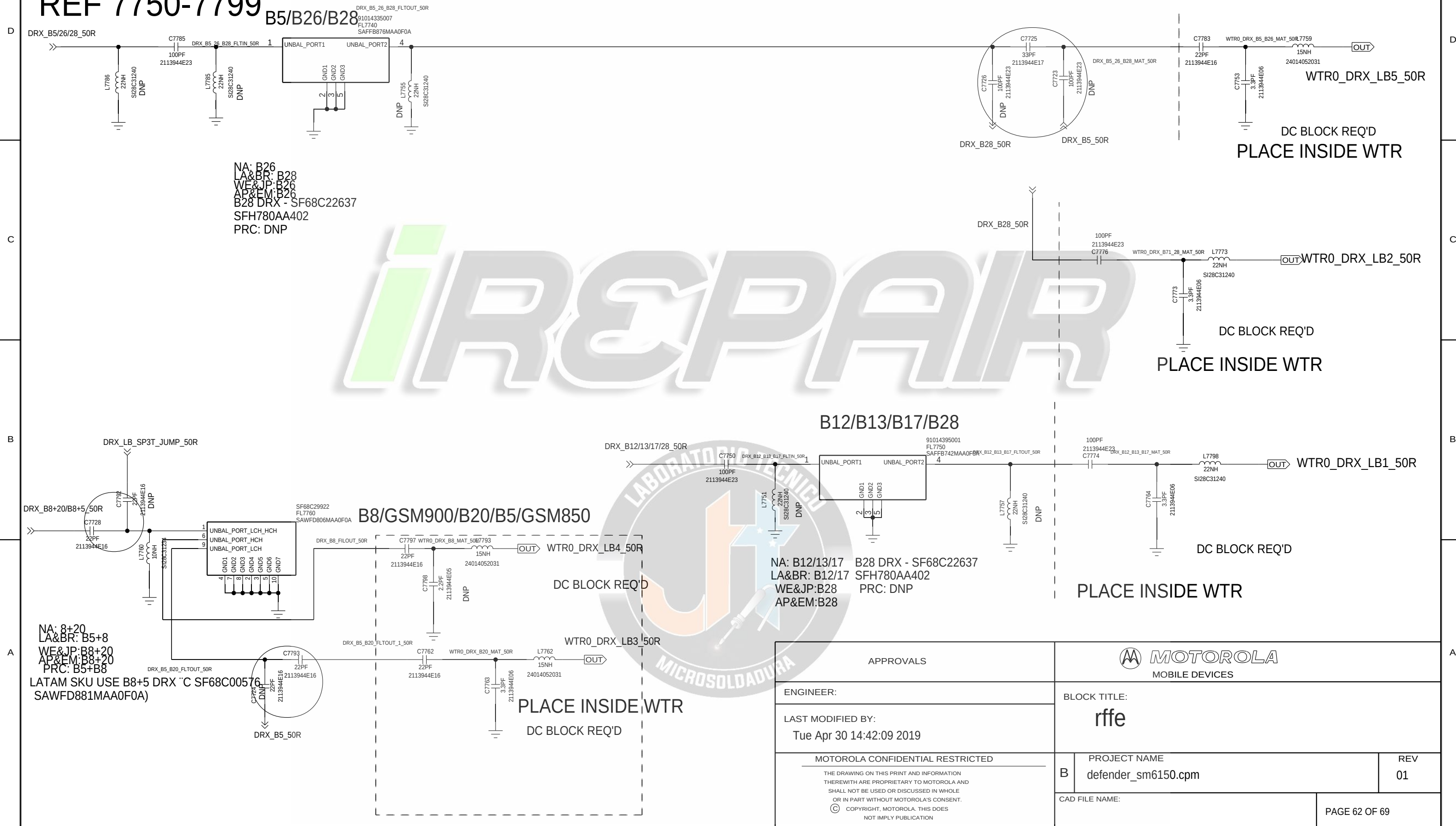
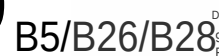
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		CAD FILE NAME:	PAGE 60 OF 69

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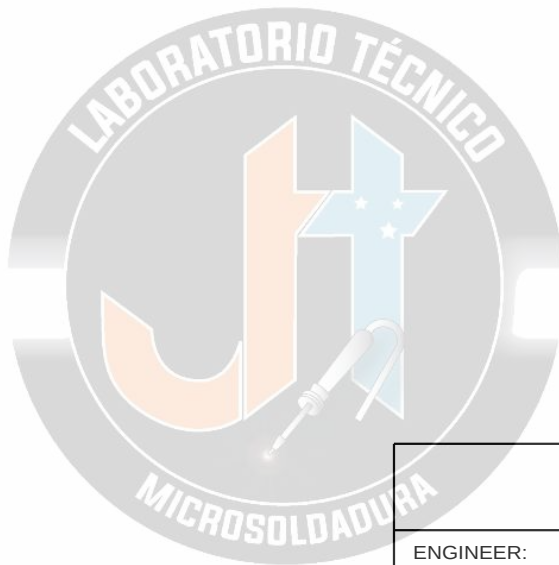
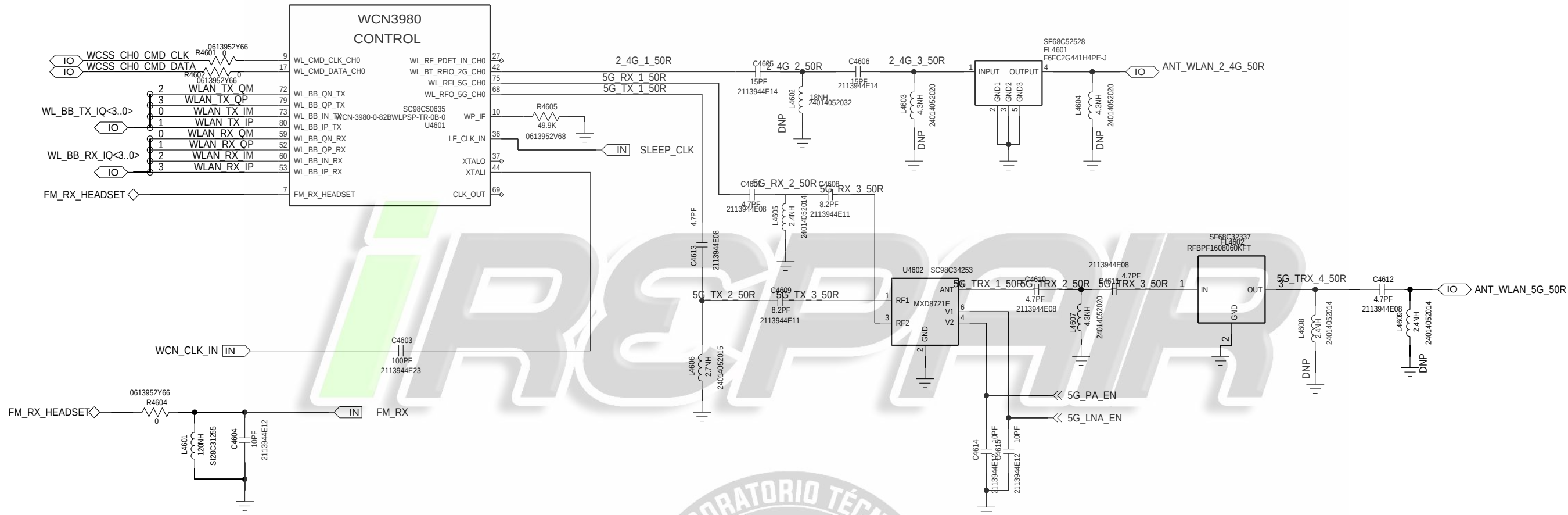


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REF 7750-7799



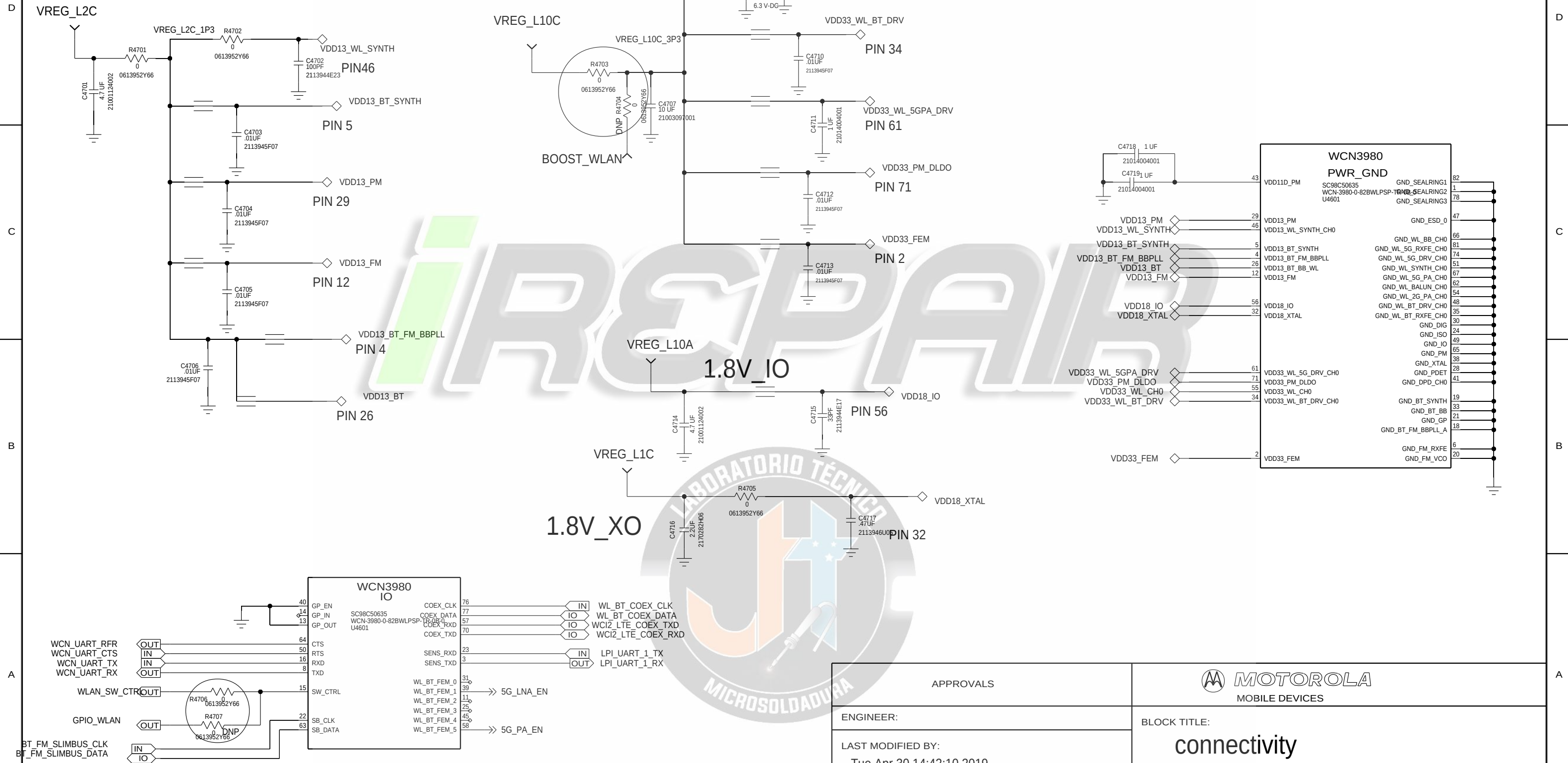
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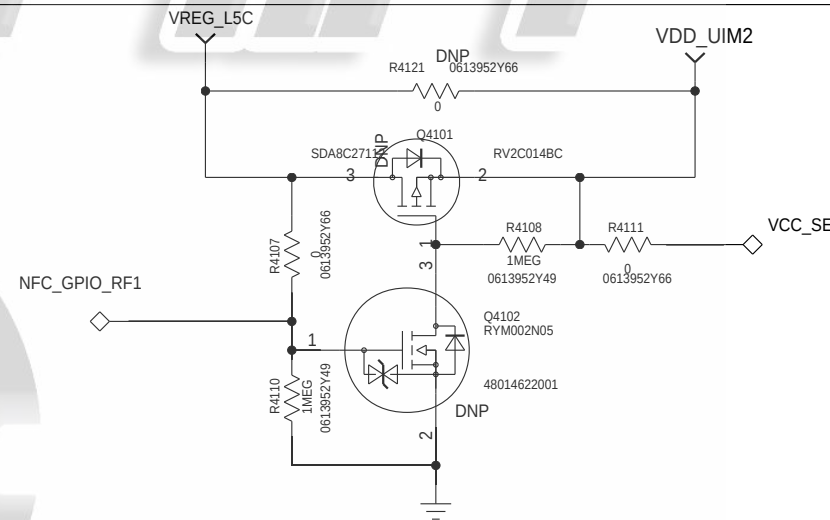
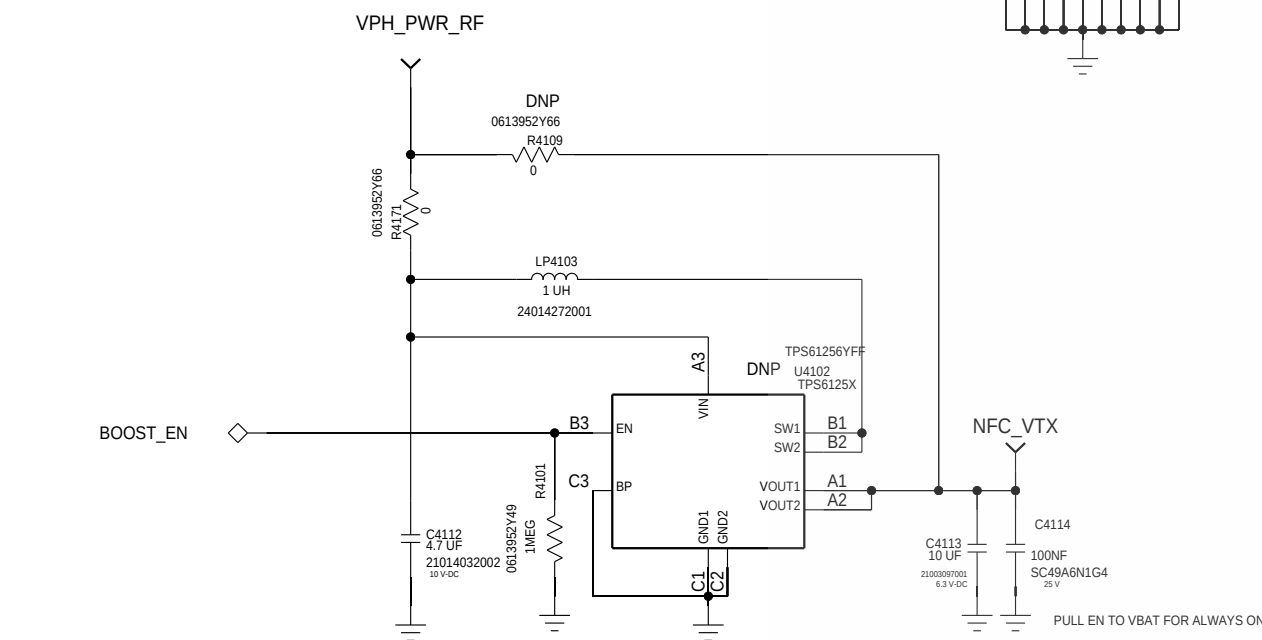
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		CAD FILE NAME:	PAGE 67 OF 69

